

Universal Portfolios

Timothy Long

June 27, 2026

Contents

1	Introduction	2
2	Portfolio Theory	2
2.1	Mathematical Framework	2
2.2	Portfolio Selection	4
2.3	The Markowitz Mean-Variance Approach	5
3	The Log-Optimal Portfolio	7
3.1	Definition	7
3.2	Notions of Optimality	9
4	Approximating Log-Optimal Investment	14
4.1	The Empirical Bayes Portfolio	17
4.2	The Universal Portfolio	21
5	Implementing the Universal Portfolio	26
5.1	Case Study	27
6	Duality between Investment and Data Compression	29
7	Conclusion	30

1 Introduction

Optimal investment in the stock market has long been an active area of research. Many see the problem of portfolio selection as dependent on the investor's utility for wealth, but we argue the existence of a universally-optimal investment scheme. The portfolio that maximises the expected logarithmic return was individually proposed by Williams [26], Kelly [7] and Latané [27]. The short-term and long-term properties that make this portfolio so attractive will be explored in §3. Let S_n and S_n^* be the wealth after n periods using an arbitrary, causal portfolio and the log-optimal portfolio respectively. We will prove Breiman's [11] result that, in the long run, the growth rate of S_n cannot exceed that of S_n^* .

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n}{S_n^*} \right) \leq 0 \right) = 1 \quad (1.1)$$

Bell and Cover's [5, 12] discovery of short-term optimality may be surprising considering the log-optimal portfolio was defined only by its long-term growth. We analyse the general zero-sum game with payoff $\mathbb{E}[\phi(W^{(1)}S_n^{(1)}/W^{(2)}S_n^{(2)})]$ to player 1 where $W^{(1)}$ and $W^{(2)}$ are fair randomisations of player 1 and player 2's initial wealth. This payoff function encompasses a number of desirable investment objectives. To play optimally in this game, it will be shown that the investor should use the log-optimal portfolio and randomise their wealth based on ϕ only.

The deep relationship between investment and data compression is discussed, as previously explored by Kelly [7] and Cover [8, 20, 21, 25]. We relate Shannon code to logarithmic investment as the optimal strategies for their respective problems. In particular, §4 finds the existence of universal portfolios (parallel to universal data compression schemes) that perform as well as if the distribution of asset returns was known but with no such assumption. Since no distribution exists in reality, this is a significant step towards an implementable investment strategy. The optimal benchmark for comparison here is the best constant rebalanced portfolio chosen with foreknowledge of the stock returns. This is exactly the log-optimal portfolio using the empirical distribution of the stock returns. Although this clairvoyant target is seemingly impossible to attain, we show that the universal portfolios achieve the same growth rate in the long run.

These universal portfolios are rebalanced daily. In an attempt to prove the validity of logarithmic investment in real markets, the impact of transaction costs has been studied by Blum and Kalai [19], Cover [28] and Iyengar [28, 29]. Our analysis shall focus on transaction costs that are proportional to the amount of wealth displaced during rebalancing. We test this effect numerically with a Monte Carlo approximation to a universal portfolio of two stocks. Under a realistic commission of 0.1%, the strategy significantly outperforms both of the underlying stocks. Rebalancing less often is found to be a successful method of reducing commission costs without harming the universal properties too much.

2 Portfolio Theory

2.1 Mathematical Framework

Consider an investor managing a portfolio with m stocks.

Definition 2.1.1. (*Portfolio vector*) The proportion of wealth invested in each of the m stocks is given by the elements of a portfolio vector, $\mathbf{b} \in \mathbb{R}^m$.

Definition 2.1.2. (*Portfolio simplex*) The space of all possible wealth allocations is the $(m-1)$ -dimensional

portfolio simplex, $\mathcal{B}$.

$$\mathcal{B} := \left\{ \mathbf{b} \in \mathbb{R}^m : \sum_i b_i = 1, b_i \geq 0 \right\} \quad (2.1)$$

The condition of $b_i \geq 0$ reflects an assumption of no short selling. Points on the interior of this simplex reflect diversified portfolios with non-zero wealth allocated to each stock. The vertices correspond to portfolios with all wealth in just one stock.

Proposition 2.1.1. *The portfolio simplex, $\mathcal{B}$, is a compact and convex set.*

Proof. $\mathcal{B}$ is compact iff $\mathcal{B}$ is closed and bounded.

The boundary of the space consists of portfolios with money allocated in $n \in \mathbb{N} < m$ stocks. Therefore, $\mathcal{B}$ is closed.

The boundedness arises from the requirement that $0 \leq b_i \leq 1$.

Let $\lambda \in [0, 1]$. Consider the weighted average of two portfolios $\mathbf{b}, \mathbf{b}' \in \mathcal{B}$.

$$\mathbf{w} := \lambda \mathbf{b} + (1 - \lambda) \mathbf{b}'$$

w_i are trivially non-negative since $b_i \geq 0$ and $\lambda \in [0, 1]$.

$$\begin{aligned} \sum_i w_i &= \sum_i (\lambda b_i + (1 - \lambda) b'_i) \\ &= \lambda \sum_i b_i + (1 - \lambda) \sum_i b'_i \\ &= \lambda + (1 - \lambda) \\ &= 1 \end{aligned}$$

Thus $\mathbf{w} \in \mathcal{B}$ and $\mathcal{B}$ is convex. □

Definition 2.1.3. (*Price relative*) The factor by which the i th stock price increases shall be called the price relative, $X_i \geq 0$.

For instance, if the i th stock doubles in price during a given period of time, $X_i = 2$.

The random state of the stock market at a given time is represented by the collection of price relatives: $\mathbf{X} = (X_1, \dots, X_m) \in \mathbb{R}^m$. Generally, the joint distribution of $\mathbf{X}$ is assumed to be known and given by $F(\mathbf{x})$. In reality, this is not the case and we can relax this assumption to achieve stronger results. We shall see that §4.1 removes the assumption that the distribution is known and instead aims to approximate it. Improving on this further, §4.2 removes the assumption that a distribution even exists.

Definition 2.1.4. (*Wealth relative*) The factor by which the wealth, invested into a portfolio $\mathbf{b}$, increases shall be called the wealth relative, S . This is identical to the wealth of the portfolio when the initial wealth is normalised to 1.

$$S := \mathbf{b}^T \mathbf{X} = \sum_{i=1}^m b_i X_i \quad (2.2)$$

Imagine the investor trades over n periods. The investor witnesses a sequence of stock market vectors: $\mathbf{X}_1, \dots, \mathbf{X}_n$. If these random vectors were i.i.d., an optimal strategy today would be an optimal strategy tomorrow.

Definition 2.1.5. (*Constant rebalanced portfolio*) A portfolio vector independent of time, $\mathbf{b}(t) = \mathbf{b}$, is called a constant rebalanced portfolio because, as the stock prices vary with time, the investor must continue to rebalance the growing wealth back into the original proportions dictated by $\mathbf{b}$.

Markets with i.i.d. $\mathbf{X}_1, \dots, \mathbf{X}_n$ have constant rebalanced optimal portfolios.

In real financial markets, the i.i.d. assumption does not hold. For instance, Mandelbrot [3] noticed volatility clustering in which “large changes [in price during period k] tend to be followed by large changes [in price during period $k + 1$]”. Consequently, we have dependence between $\mathbf{X}_1, \dots, \mathbf{X}_n$ and a varying distribution due to the non-constant variance.

To obtain further generality, the investor may be allowed to adapt the portfolio based on the information currently available.

Definition 2.1.6. (*Causal portfolio*) A causal portfolio is an investment strategy that is independent of the future stock market vectors.

Thus, using any causal strategy, the factor by which the wealth increases over n periods is given by:

$$S_n = \prod_{i=1}^n \mathbf{b}_i^T (\mathbf{X}_1, \dots, \mathbf{X}_{i-1}) \mathbf{X}_i. \quad (2.3)$$

2.2 Portfolio Selection

The problem of portfolio selection relates to determining a portfolio that satisfies certain investment objectives. The exact objectives have long been debated but typically focus on maximising return and minimising risk.

Consider a function, $f(\mathbf{b})$, that represents the investment objective we wish to maximise. Let the optimal portfolio be the $\mathbf{b}$ that attains this maximum.

Remark 2.2.1. $f(\mathbf{b})$ can be thought of as the expected utility of the portfolio, $\mathbf{b}$, to the investor.

$$f(\mathbf{b}) = \mathbb{E}[U(\mathbf{b})] \quad (2.4)$$

Theorem 2.2.1. (*Well-posedness of portfolio selection*) Portfolio selection based on maximising $f(\mathbf{b})$ is a well-posed problem if $f(\mathbf{b})$ is continuous and strictly concave with respect to $\mathbf{b}$.

Proof. If $f(\mathbf{b})$ is continuous, the extreme value theorem guarantees that the maximum is attained since the portfolio simplex is compact. In other words, a portfolio that is optimal with respect to the given investment objective exists.

Strict concavity of $f(\mathbf{b})$ ensures that the optimal portfolio that attains this maximum is unique.

Stability of the problem is guaranteed by the continuity of $f(\mathbf{b})$. Small deviations in f lead to small deviations in the optimal portfolio calculated. $\square$

The optimal portfolio is the solution to the following convex optimisation problem.

$$\mathbf{b}_{opt} := \arg \max_{\mathbf{b} \in \mathcal{B}} f(\mathbf{b}) \quad (2.5)$$

Methods of portfolio selection simply differ on the choice of this function, $f(\mathbf{b})$.

2.3 The Markowitz Mean-Variance Approach

The naïve investor may choose to maximise the expected portfolio return, $f(\mathbf{b}) = \mathbb{E}[\mathbf{b}^T \mathbf{X}]$. This financial objective fails to consider the risk of the portfolio and suggests the investor has a linear utility for wealth.

Proposition 2.3.1. *The financial objective $f(\mathbf{b}) = \mathbb{E}[\mathbf{b}^T \mathbf{X}]$ does not favour diversification.*

Proof. The possible optimal portfolios are given by:

$$\begin{aligned} \mathbf{b}_{opt} &\in \arg \max_{\mathbf{b} \in \mathcal{B}} \mathbb{E}[\mathbf{b}^T \mathbf{X}] \\ &\in \arg \max_{\mathbf{b} \in \mathcal{B}} \sum_{i=1}^m b_i \mathbb{E}[X_i]. \end{aligned}$$

Due to the constraints $b_i \geq 0$ and $\sum_i b_i = 1$, the optimal portfolio would simply be placing all wealth into the stock(s) with the largest $\mathbb{E}[X_i]$. This process is not necessarily unique.

Suppose we trade three stocks with the following expected price relatives.

$$\mathbb{E}[X_1] = \mathbb{E}[X_2] = 1.5 > 1.1 = \mathbb{E}[X_3]$$

Then the following portfolios are equally optimal.

$$\mathbf{b}_1 = (1, 0, 0), \quad \mathbf{b}_2 = (0, 1, 0), \quad \mathbf{b}_{12} = (0.5, 0.5, 0)$$

Thus, the financial objective $f(\mathbf{b}) = \mathbb{E}[\mathbf{b}^T \mathbf{X}]$ does not reward diversified portfolios. □

Markowitz argues that “a rule of behaviour which does not imply the superiority of diversification must be rejected” [1]. This motivates the mean-variance approach to portfolio selection. Markowitz uses the variance of the portfolio’s return as a measure of the portfolio’s risk. The method suggests an investor should aim to maximise expected return and minimise the variance of the return.

Definition 2.3.1. (Efficient portfolio) *Let a portfolio, $\mathbf{b} \in \mathcal{B}$, have expected return $\mathbb{E}[\mathbf{b}^T \mathbf{X}]$ and variance of return $V[\mathbf{b}^T \mathbf{X}]$. This portfolio is efficient if and only if there does not exist a portfolio with a greater expected return and an equal or smaller variance and there does not exist a portfolio with a smaller variance and an equal or greater expected return.*

There is a set of efficient portfolios from which an investor may select a portfolio that best matches their investment objectives. For instance, an investor may select the efficient portfolio with low variance (at a cost of expected return) for a safer, long-term investment strategy.

A robust formulation of mean-variance portfolio selection writes the financial objective in the form of a regularised optimisation problem.

$$f(\mathbf{b}) = \mathbb{E}[\mathbf{b}^T \mathbf{X}] - \lambda V[\mathbf{b}^T \mathbf{X}] \tag{2.6}$$

$\lambda > 0$ controls the trade-off between the risk and the return of the portfolio [2]. The risk-averse investor selects a larger λ .

Proposition 2.3.2. *Mean-variance portfolio selection, attained by maximising the objective in (2.6) over the simplex, is well-posed.*

Proof.

$$\begin{aligned} f(\mathbf{b}) &= \mathbb{E}[\mathbf{b}^T \mathbf{X}] - \lambda V[\mathbf{b}^T \mathbf{X}] \\ &= \mathbf{b}^T \mathbb{E}[\mathbf{X}] - \lambda \mathbf{b}^T V[\mathbf{X}] \mathbf{b} \end{aligned}$$

Write $\mathbb{E}[\mathbf{X}]$ as the mean stock market vector, $\mu \in \mathbb{R}^m$, and $V[\mathbf{X}]$ as the covariance matrix, $\Sigma \in \mathbb{R}^{m \times m}$, of the stock market vector.

$$f(\mathbf{b}) = \mathbf{b}^T \mu - \lambda \mathbf{b}^T \Sigma \mathbf{b}$$

This function is continuous everywhere with respect to $\mathbf{b}$.

One can check for strict concavity by constructing the Hessian matrix.

$$\begin{aligned} \frac{\partial f(\mathbf{b})}{\partial \mathbf{b}^T} &= \mu - 2\lambda \Sigma \mathbf{b} \\ \mathbf{H} &= \frac{\partial^2 f(\mathbf{b})}{\partial \mathbf{b} \partial \mathbf{b}^T} = -2\lambda \Sigma \end{aligned}$$

$\lambda > 0$ is required so that we improve on the ill-posed financial objective $f(\mathbf{b}) = \mathbb{E}[\mathbf{b}^T \mathbf{X}]$ previously discussed. By definition, all covariance matrices are positive semi-definite, $\Sigma \succeq 0$. If Σ was singular, the variance of the return would be zero, $\mathbf{b}^T \Sigma \mathbf{b} = V[\mathbf{b}^T \mathbf{X}] = 0$. Thus we assume a non-singular covariance matrix, again to stray from the case of maximising expected return alone. Therefore, the covariance matrix is positive definite, $\Sigma \succ 0$, and the Hessian is negative definite, $\mathbf{H} \prec 0$. Negative definiteness of the Hessian implies $f(\mathbf{b})$ is strictly concave.

By Theorem 2.2.1, the strict concavity and continuity of $f(\mathbf{b})$ ensures this problem is well-posed. $\square$

By penalising the variance, diversification is encouraged with this method of portfolio selection. The Markowitz approach is also praised for its simplicity; finding the optimal portfolio ultimately comes down to determining the mean and variance of the stock market vector.

Cover and Bell [5] point out the disadvantages to using only two moments (the mean and variance) of the price relative distribution. A normal distribution is completely specified by these two moments but Mandelbrot [3] and Fama [4] criticise the assumption that returns are normally distributed. Empirical data suggests the distribution of returns is leptokurtic with extreme returns far more common than a normal distribution would suggest.

Another downside to mean-variance portfolio selection is the method's inability to identify a single optimal portfolio upon which all investors agree. Individual investors must still choose between efficient portfolios or choose the regularisation parameter, λ .

Finally, Hanoch and Levy [6] disagree with the classification of risk using variance. The financial objective in (2.6) penalises upside and downside variance symmetrically but a rational investor would prefer upside variance. This can be attributed to the implied utility function's failure to be monotonically increasing. As a result, Hanoch and Levy [6] were able to provide scenarios, using alternative utility functions, in which the optimal portfolio was not in the set of efficient portfolios.

These failures motivate the following introduction of an alternative optimal portfolio: the log-optimal portfolio.

3 The Log-Optimal Portfolio

3.1 Definition

As investors tend to reinvest their capital, wealth compounds multiplicatively over time. For a constant rebalanced portfolio $\mathbf{b}$, the wealth relative after n periods is:

$$S_n = \prod_{i=1}^n \mathbf{b}^T \mathbf{X}_i \quad (3.1)$$

where $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. according to the known price relative distribution, $F(\mathbf{x})$.

In order to confidently describe the behaviour of an investment strategy in the long term, we require the use of the strong law of large numbers. This guarantees the (almost sure) convergence of our actual portfolio performance to an intended investment objective in the long run. The logarithm converts the product in (3.1) to a summation where we can apply the law of large numbers.

$$\frac{1}{n} \log(S_n) = \frac{1}{n} \sum_{i=1}^n \log(\mathbf{b}^T \mathbf{X}_i) \quad (3.2)$$

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \log(S_n) = \mathbb{E}[\log(\mathbf{b}^T \mathbf{X})] \right) = 1 \quad (3.3)$$

Kelly [7] introduced the notion of maximising the expected logarithmic return to play optimally in a game where the bettor has access to unfair information through a noisy channel. The paper found a parallel between maximising the rate at which wealth grows and maximising the rate of information through the channel.

Definition 3.1.1. (*Doubling rate [8]*) The expected value of the logarithm of the wealth relative is called the doubling rate, $W(\mathbf{b})$ ¹.

$$W(\mathbf{b}) := \mathbb{E}[\log(\mathbf{b}^T \mathbf{X})] \quad (3.4)$$

Proposition 3.1.1. Portfolio selection based on maximising the financial objective $f(\mathbf{b}) = W(\mathbf{b})$ is well-posed.

Proof. Let $f(\mathbf{b}) = W(\mathbf{b}) = \mathbb{E}[\ln(\mathbf{b}^T \mathbf{X})]$. Although our doubling rate was defined using base 2 logarithms, we shall revert back to natural logarithms here for simplicity with the derivatives. No generality is lost here as we just pick up a (positive) scalar factor via the base change rule of logarithms.

We shall check for strict concavity by considering the Hessian.

$$\begin{aligned} \frac{\partial f(\mathbf{b})}{\partial \mathbf{b}^T} &= \mathbb{E} \left[\frac{\mathbf{X}}{\mathbf{b}^T \mathbf{X}} \right] \\ \mathbf{H} &= \frac{\partial^2 f(\mathbf{b})}{\partial \mathbf{b} \partial \mathbf{b}^T} = \mathbb{E} \left[\frac{-\mathbf{X} \mathbf{X}^T}{(\mathbf{b}^T \mathbf{X})^2} \right] \end{aligned}$$

Trivially, $(\mathbf{b}^T \mathbf{X})^2 > 0$. Consider $\mathbb{E}[\mathbf{X} \mathbf{X}^T]$ and let $\mathbf{v} \in \mathbb{R}^m$ be arbitrary.

$$\begin{aligned} \mathbf{v}^T \mathbb{E}[\mathbf{X} \mathbf{X}^T] \mathbf{v} &= \mathbb{E}[(\mathbf{v}^T \mathbf{X})^2] \\ &\geq 0 \end{aligned}$$

¹We use the term ‘doubling rate’ here because our logarithm is base 2. This is merely a choice to further mirror the field of information theory where we measure in bits.

Therefore, $\mathbb{E}[\mathbf{X}\mathbf{X}^T]$ is positive semi-definite. If it were singular, there would exist a vector $\mathbf{v} \in \mathbb{R}^m \setminus \{0\}$ such that $\mathbf{v}^T \mathbf{X} = 0$. In other words, $\mathbb{E}[\mathbf{X}\mathbf{X}^T]$ is singular if the price relatives are linearly dependent. Assume this is not the case and we have $\mathbb{E}[\mathbf{X}\mathbf{X}^T] \succ 0$. Therefore, $\mathbf{H} \prec 0$ and the financial objective is strictly concave.

Furthermore, $\log(\mathbf{b}^T \mathbf{X})$ is continuous for all $\mathbf{b}^T \mathbf{X} > 0$.

By Theorem 2.2.1, this portfolio selection problem is well-posed. $\square$

Definition 3.1.2. (*Log-optimal portfolio*) The log-optimal portfolio, $\mathbf{b}^* \in \mathcal{B}$, is the unique maximiser of the doubling rate.

$$\mathbf{b}^* := \arg \max_{\mathbf{b} \in \mathcal{B}} W(\mathbf{b}) = \arg \max_{\mathbf{b} \in \mathcal{B}} \mathbb{E}[\log(\mathbf{b}^T \mathbf{X})] \quad (3.5)$$

Let $W^* := W(\mathbf{b}^*)$ be the optimal growth rate of wealth.

Immediately, we have the advantage that all investors agree upon a single optimal portfolio. Unlike in the Markowitz method, no further decisions regarding the risk-return trade-off are required.

Remark 3.1.1. The financial objective $W(\mathbf{b})$ depends on the entire distribution of price relatives, improving on just the two moments used by Markowitz.

Proof. Again, for simplicity with the expansion of the logarithm, we revert back to natural logarithms for this proof. No generality is lost here for the same reasons.

Consider the Taylor expansion of the logarithm here. For simplicity, write $S = \mathbf{b}^T \mathbf{X}$ and $\bar{S} = \mathbb{E}[\mathbf{b}^T \mathbf{X}]$.

$$\begin{aligned} \ln(\mathbf{b}^T \mathbf{X}) &= \ln(S) \\ &= \ln(\bar{S} + (S - \bar{S})) \\ &= \ln(\bar{S}) + \ln\left(1 + \frac{S - \bar{S}}{\bar{S}}\right) \\ &= \ln(\bar{S}) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (S - \bar{S})^n}{n \bar{S}^n} \end{aligned}$$

Taking expectation over the distribution of price relatives, we arrive at the exponential growth rate, $W(\mathbf{b})$.

$$W(\mathbf{b}) = \ln(\bar{S}) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \mathbb{E}[(S - \bar{S})^n]}{n \bar{S}^n} \quad (3.6)$$

$\mathbb{E}[(S - \bar{S})^n]$ is the n th central moment.

This series converges for $|(S - \bar{S})/\bar{S}| < 1 \implies |S - \bar{S}| < \bar{S}$. Of course, this is not valid in general, but this proof serves as a simple demonstration where the moments appear naturally. $\square$

Proposition 3.1.2. Use of the log-optimal portfolio guarantees the realised doubling rate of the investor's wealth converges almost surely to the maximum doubling rate.

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{1}{n} \log(S_n^*) = W^*\right) = 1 \quad (3.7)$$

Here, S_n^* denotes the wealth relative after n days using the log-optimal portfolio.

Proof. Simply apply the strong law of large numbers in (3.3) to the log-optimal portfolio and use the definition of the doubling rate in (3.4). $\square$

Stronger notions of optimality of the log-optimal portfolio will be discussed in §3.2.

Despite these benefits, Samuelson [9] strongly opposes the premise of logarithmic investment due to the implication that the investor has a logarithmic utility for wealth [5]. Of course, the investor is not required to have such a utility and thus, with differing utilities, logarithmic investment is shown to be sub-optimal to other strategies [9]. In addition to this, Samuelson rejects the idea of a universally-optimal portfolio considering each investor may have a unique perspective on the utility of wealth. Thorp [10] argues that the logarithmic utility should be seen as prescriptive rather than descriptive or predictive. In other words, it is a tool that leads us to various notions of optimality and doesn't aim to mirror the behaviour of investors.

3.2 Notions of Optimality

In an attempt to defend logarithmic investment, we shall collate a number of properties of the log-optimal portfolio which display the short-term, long-term and competitive benefits.

In the case of an i.i.d. market, Breiman [11] proved that investing with the log-optimal portfolio is asymptotically optimal in the sense that an investor using any other causal portfolio will almost surely have a smaller growth rate of wealth in the long run. Theorem 3.2.1 formalises this idea in the style of Cover's later proof [8].

Theorem 3.2.1. *(Asymptotic optimality [8]) Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be i.i.d. stock market vectors distributed according to $F(\mathbf{x})$. Consider an investor using an arbitrary sequence of causal portfolios with wealth relative S_n . Consider another investor using the log-optimal portfolio, $\mathbf{b}^*$, with wealth relative S_n^* after n periods. Then the following holds.*

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n}{S_n^*} \right) \leq 0 \right) = 1 \quad (3.8)$$

Proof. The log-optimal portfolio is the portfolio satisfying the optimisation problem: $\min -W(\mathbf{b})$ subject to $\sum_i b_i = 1$ and $b_i \geq 0$.

We can convert this problem into a Lagrangian.

$$\mathcal{L}(\mathbf{b}) := -W(\mathbf{b}) - \lambda \left(1 - \sum_{i=1}^m b_i \right) - \mu^T \mathbf{b}$$

$\lambda \in \mathbb{R}$ is the Lagrange multiplier ensuring the portfolio vector elements sum to 1. $\mu \in \mathbb{R}^m$ are Kuhn-Tucker multipliers (generalisations of Lagrange multipliers to inequality constraints) satisfying $\mu_i \geq 0$ for all i . When we minimise $\mathcal{L}$ with respect to $\mathbf{b}$, $-\mu^T \mathbf{b}$ is minimised when each element of the portfolio vector is positive, thus ensuring no shorting of stocks.

Consider the partial derivative of $\mathcal{L}$ with respect to the elements of the portfolio vector. Again, imagine we defined the $W(\mathbf{b})$ with respect to the natural logarithm for ease.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial b_i} &= -\frac{\partial W}{\partial b_i} + \lambda - \mu_i \\ &= -\mathbb{E} \left[\frac{X_i}{\mathbf{b}^T \mathbf{X}} \right] + \lambda - \mu_i \end{aligned}$$

The log-optimal portfolio minimises this Lagrangian.

$$\mathbb{E} \left[\frac{X_i}{\mathbf{b}^{*T} \mathbf{X}} \right] = \lambda - \mu_i$$

To evaluate λ , multiply this equation by b_i^* and sum over all i . Using $\sum_i b_i^* = 1$ we find:

$$\sum_{i=1}^m b_i^* \mathbb{E} \left[\frac{X_i}{\mathbf{b}^{*T} \mathbf{X}} \right] = \lambda - \sum_{i=1}^m \mu_i b_i^*.$$

Via the linearity of expectation:

$$\begin{aligned} \mathbb{E} \left[\frac{\mathbf{b}^{*T} \mathbf{X}}{\mathbf{b}^{*T} \mathbf{X}} \right] &= \lambda - \sum_{i=1}^m \mu_i b_i^* \\ 1 &= \lambda - \sum_{i=1}^m \mu_i b_i^*. \end{aligned}$$

The Kuhn-Tucker conditions state that $\mu^T \mathbf{b}^* = 0$. Therefore, $\lambda = 1$ and the log-optimal portfolio satisfies the necessary condition:

$$\mathbb{E} \left[\frac{X_i}{\mathbf{b}^{*T} \mathbf{X}} \right] = 1 - \mu_i \leq 1.$$

Consider another investor using an arbitrary portfolio, $\mathbf{b}$, with wealth S .

$$\mathbb{E} \left[\frac{S}{S^*} \right] \leq 1$$

This analysis may be extended to an investor using a causal portfolio over n periods.

$$\mathbb{E} \left[\frac{S_n}{S_n^*} \right] \leq 1$$

S_n/S_n^* is a non-negative random variable and Markov's inequality may be used.

$$\begin{aligned} \mathbb{P} \left(\frac{S_n}{S_n^*} \geq t_n \right) &\leq \frac{1}{t_n} \mathbb{E} \left[\frac{S_n}{S_n^*} \right] \\ &\leq \frac{1}{t_n} \end{aligned}$$

Due to the increasing nature of the logarithm function, this can be rewritten as:

$$\mathbb{P} \left(\frac{1}{n} \log \left(\frac{S_n}{S_n^*} \right) \geq \frac{1}{n} \log(t_n) \right) \leq \frac{1}{t_n}$$

where $t_n > 0$ is a function of n that may be chosen.

The Borel-Cantelli lemma states that if $\sum_n \mathbb{P}(A_n) < \infty$ then $\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0$. Many choices of t_n lead us to the desired form; Cover [8] uses $t_n = n^2$ for instance.

We will choose $t_n = e^{n\varepsilon}$, for any $\varepsilon > 0$, so that $\frac{1}{n} \log(t_n)$ reduces to ε .

$$\sum_{n=1}^{\infty} \frac{1}{t_n} = \sum_{n=1}^{\infty} \frac{1}{e^{n\varepsilon}}$$

Test convergence of this series using the ratio test.

$$\begin{aligned} L &:= \lim_{n \rightarrow \infty} \left| \frac{e^{n\varepsilon}}{e^{(n+1)\varepsilon}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{1}{e^\varepsilon} \right| \\ &< 1 \end{aligned}$$

Since $L < 1$, we have convergence and we may use the Borel-Cantelli lemma.

$$\begin{aligned} \mathbb{P} \left(\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n}{S_n^*} \right) \geq \limsup_{n \rightarrow \infty} \frac{1}{n} \log(e^{n\varepsilon}) \right) &= 0 \\ \mathbb{P} \left(\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n}{S_n^*} \right) \geq \varepsilon \right) &= 0 \end{aligned}$$

As this is true for any $\varepsilon > 0$, we arrive at the expression for asymptotic optimality of the log-optimal portfolio.

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n}{S_n^*} \right) \leq 0 \right) = 1$$

□

Breiman [11] also proved that the log-optimal portfolio minimises the time taken for the wealth to reach a pre-defined amount, A , in the limit as $A \rightarrow \infty$.

Theorem 3.2.2. (*Asymptotic time minimisation*) Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be i.i.d. stock market vectors distributed according to $F(\mathbf{x})$. Say an investor uses an arbitrary sequence of causal portfolios with wealth relative S_n . Consider the time taken to reach a wealth relative of $A \in \mathbb{R}^+$.

$$T(A) := \inf\{n \in \mathbb{N} : S_n \geq A\} \quad (3.9)$$

The log-optimal portfolio is the unique minimiser of $\mathbb{E}[T(A)]$ asymptotically as $A \rightarrow \infty$.

Proof. For simplicity, we shall prove the log-optimal portfolio has an $\mathbb{E}[T(A)]$ less than or equal to that of any other constant rebalanced portfolio, as opposed to the more general case of a causal portfolio. A comprehensive proof is done by Breiman [11].

The wealth relative after n days with a constant rebalanced portfolio is given by:

$$S_n = \prod_{i=1}^n \mathbf{b}^T \mathbf{X}_i.$$

Let $N := T(A)$ be the first period in which this arbitrary causal portfolio has a wealth relative greater than or equal to A . Define S_N precisely in terms of a non-negative error term, ε , representing the amount that $\log(S_N)$ exceeds $\log(A)$.

$$\begin{aligned} \log(S_N) &:= \log(A) + \varepsilon \\ \sum_{i=1}^N \log(\mathbf{b}^T \mathbf{X}_i) &= \log(A) + \varepsilon \\ \mathbb{E} \left[\sum_{i=1}^N \log(\mathbf{b}^T \mathbf{X}_i) \right] &= \log(A) + \mathbb{E}[\varepsilon] \end{aligned}$$

Under our assumption that the stock market vectors, $\mathbf{X}_1, \dots, \mathbf{X}_n$, are i.i.d. random variables then $\log(\mathbf{b}^T \mathbf{X}_1), \dots, \log(\mathbf{b}^T \mathbf{X}_n)$ are too. Assuming finite doubling rate and finite $\mathbb{E}[T(A)]$, Wald's identity tells us:

$$\mathbb{E} \left[\sum_{i=1}^N \log(\mathbf{b}^T \mathbf{X}_i) \right] = \mathbb{E}[N] \mathbb{E}[\log(\mathbf{b}^T \mathbf{X})].$$

Substitute $T(A)$ and $W(\mathbf{b})$ in.

$$\mathbb{E}[T(A)] = \frac{\log(A)}{W(\mathbf{b})} + \frac{\mathbb{E}[\varepsilon]}{W(\mathbf{b})}$$

In the limit as $A \rightarrow \infty$, the $\mathbb{E}[T(A)]$ and $\log(A)$ terms dominate and we can neglect the constant expected value of this error term, $\mathbb{E}[\varepsilon]$.

Under the condition $A \rightarrow \infty$, the expected time to reach a capital A may be minimised by maximising the doubling rate which is accomplished through use of the log-optimal portfolio. $\square$

Theorems 3.2.1 and 3.2.2 focus on the asymptotic optimality of the log-optimal portfolio in the long run. These properties are a natural consequence of maximising the growth rate of wealth. However, investors don't witness an infinite number of time periods and there is no guarantee that the log-optimal portfolio performs well in short-term investment.

Bell and Cover [5, 12] investigated the short-term behaviour through the study of zero-sum games where an investor aims to outperform another in any number of time periods.

Consider the two-player, zero-sum game where the payoff to player 1 is $\mathbb{E}[\phi(S_n^{(1)}/S_n^{(2)})]$ and $S_n^{(1)}$ and $S_n^{(2)}$ are the wealth relatives of player 1 and 2 after n periods. $\mathbb{E}[\phi(S_n^{(1)}/S_n^{(2)})]$ is a generalised payoff function that should encompass a large number of possible definitions of success for an investor. Since the payoff to player 1 should increase with $S_n^{(1)}/S_n^{(2)}$, ϕ is non-decreasing. The game is played by selecting a causal portfolio sequence.

Suppose both players begin with the same initial wealth. Due to the short-term nature of the game, certain choices of ϕ allow a player to gain an edge by using a fair gamble with an ε chance of zero wealth and a $1 - \varepsilon$ chance of beginning the game with greater wealth than their opponent [5]. This strategy requires knowledge of the opponent's starting wealth and so game theory suggests starting the game with a randomisation depending only on ϕ . Bell and Cover [12] provide a deeper analysis describing the conditions on ϕ for the randomisation to be necessary.

Definition 3.2.1. (Fair gamble) A fair gamble of the initial wealth is a random variable, $W \sim G$, such that $W \geq 0$ and $\mathbb{E}[W] \leq 1$. In other words, a gamble is fair if it can't lead to negative wealth and is not expected to increase the starting wealth.

Definition 3.2.2. (Primitive ϕ -game [12]) The primitive ϕ -game is a two-player game with payoff $\mathbb{E}[\phi(W^{(1)}/W^{(2)})]$ to player 1. Each player chooses a fair gamble of their starting wealth, $W^{(1)} \sim G^{(1)}$ and $W^{(2)} \sim G^{(2)}$, in an attempt to outperform their opponent.

The payoff, $\mathbb{E}[\phi(W^{(1)}/W^{(2)})]$, to player 1 can be expressed as:

$$\mathbb{E} \left[\phi \left(\frac{W^{(1)}}{W^{(2)}} \right) \right] = \int \int \phi \left(\frac{w^{(1)}}{w^{(2)}} \right) dG^{(1)}(w^{(1)}) dG^{(2)}(w^{(2)}). \quad (3.10)$$

The optimal distributions in the primitive ϕ -game, $G_*^{(1)}$ and $G_*^{(2)}$, lie on a saddle point. If player 1 diverts from this optimal distribution, $G_*^{(1)}$, then the payoff to player 1, $\mathbb{E}[\phi(W^{(1)}/W^{(2)})]$, should decrease. Equally, if player 2 diverts from this optimal distribution, $G_*^{(2)}$, then the payoff to player 1 should increase. For all fair distribution functions, $G^{(1)}$ and $G^{(2)}$, this minimax condition that describes the optimal distributions can be written as [12]:

$$\int \int \phi dG^{(1)} dG_*^{(2)} \leq \int \int \phi dG_*^{(1)} dG_*^{(2)} \leq \int \int \phi dG_*^{(1)} dG^{(2)}. \quad (3.11)$$

Note that the optimal distributions depend solely on the definition of success, ϕ .

We can now adapt our game to accommodate for randomisations of the initial wealth.

Definition 3.2.3. (Stock market ϕ -game [12]) Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be i.i.d. stock market vectors distributed according to $F(\mathbf{x})$. In the stock market ϕ -game, two investors choose a fair gamble of their starting wealth, $W^{(1)} \sim G^{(1)}$ and $W^{(2)} \sim G^{(2)}$, followed by a choice of causal portfolio, $\{\mathbf{b}_i^{(1)}\}$ and $\{\mathbf{b}_i^{(2)}\}$ for $i \in \{1, \dots, n\}$, in order to outperform their opponent. The payoff to player 1 is given by $\mathbb{E}[\phi(W^{(1)}S_n^{(1)}/W^{(2)}S_n^{(2)})]$ where $S_n^{(1)}$ and $S_n^{(2)}$ are the wealth relatives of each player after n periods.

Theorem 3.2.3. (Competitive optimality [5, 12, 8]) The optimal randomisation for the stock market ϕ -game is the solution of the primitive ϕ -game, $W_* \sim G_*$. The optimal choice of portfolio in the stock market ϕ -game is the log-optimal portfolio, $\mathbf{b}^*$.

Proof. Say player 1 uses the log-optimal portfolio with wealth relative S_n^* and a fair randomisation $W_* \sim G_*$ that solves the primitive ϕ -game. Let player 2 use an arbitrary causal portfolio with wealth relative S_n and a fair randomisation $W \sim G$.

Then the Kuhn-Tucker conditions (as seen in the proof of Theorem 3.2.1) tell us:

$$\mathbb{E} \left[\frac{S_n}{S_n^*} \right] \leq 1.$$

Furthermore, $S_n/S_n^* \geq 0$ which means that S_n/S_n^* is a fair gamble. As well as this, the randomisation W is independent of the stock market vectors.

$$\mathbb{E} \left[\frac{WS_n}{S_n^*} \right] = \mathbb{E}[W] \mathbb{E} \left[\frac{S_n}{S_n^*} \right] \leq 1$$

Again, $WS_n/S_n^* \geq 0$ so $W' := WS_n/S_n^*$ is a fair gamble. This means the payoff of the stock market ϕ -game to player 1 may be simplified as follows.

$$\mathbb{E} \left[\phi \left(\frac{W_* S_n^*}{WS_n} \right) \right] = \mathbb{E} \left[\phi \left(\frac{W_*}{W'} \right) \right]$$

In other words, this game has reduced to the primitive ϕ -game and, by definition of W_* , player 1 is playing optimally here and can't be beaten by player 2.

$$\mathbb{E} \left[\phi \left(\frac{W_* S_n^*}{WS_n} \right) \right] \geq V_\phi$$

V_ϕ is the value of the primitive ϕ -game which is the payoff when both players use the optimal distribution function.

Now reverse the roles of player 1 and player 2.

$$\mathbb{E} \left[\phi \left(\frac{WS_n}{W_* S_n^*} \right) \right] = \mathbb{E} \left[\phi \left(\frac{W'}{W_*} \right) \right] \leq V_\phi$$

Therefore, player 2 is playing optimally (by decreasing the payoff to player 1) when they use the log-optimal portfolio and a fair randomisation that solves the primitive ϕ -game.

This analysis leads us to the minimax conditions seen in (3.11). There is a saddle point when both players use this specified strategy which indicates that this is indeed the optimal strategy for the stock market ϕ -game. We see that the value of this stock market ϕ -game is simply the value of the primitive ϕ -game, V_ϕ . $\square$

Remark 3.2.1. A convenient choice of $\phi(S^{(1)}/S^{(2)})$ is $\mathbb{1}\{S^{(1)}/S^{(2)} \geq 1\}$ which leads to an easily-interpreted payoff of $\mathbb{P}(S^{(1)} \geq S^{(2)})$ for player 1. This was the original payoff explored by Bell and Cover [5] for one time period.

Theorem 3.2.3 establishes the short-term optimality of the log-optimal portfolio. Use of $\mathbf{b}^*$ (along with a randomisation of the initial wealth in certain cases) ensures the investor is expected to outperform any rival investor in a finite number of time periods, $n \geq 1$. Peculiarly, in maximising the expected logarithmic return to guarantee asymptotic performance, we have also found short-term optimality.

As of yet, our analysis has been restricted to i.i.d. stock market vectors, $\mathbf{X}_1, \dots, \mathbf{X}_n$. As the stock market has no dependence on the past (no memory), the optimal portfolio at one time should naturally be the optimal portfolio at any time.

We seek a less restrictive framework for our stock market vectors in which we can still employ the strong law of large numbers to have some assurance of optimal behaviour in the long run. Algoet and Cover [13] improved Breiman's [11] i.i.d. assumption by taking the market to be stationary and ergodic. The above theorems of optimality can be extended to this model by replacing the constant rebalanced portfolio, $\mathbf{b}^*$, with a causal sequence of conditionally log-optimal portfolios, $\{\mathbf{b}_i^*(\mathbf{X}_1, \dots, \mathbf{X}_{i-1})\}$ for $i \in \mathbb{N}$.

$$\mathbf{b}_i^*(\mathbf{X}_1, \dots, \mathbf{X}_{i-1}) := \arg \max_{\mathbf{b} \in \mathcal{B}} \mathbb{E}[\log(\mathbf{b}^T \mathbf{X}_i) | \mathbf{X}_1, \dots, \mathbf{X}_{i-1}] \quad (3.12)$$

The ergodic theorem provides an equivalent form of the strong law of large numbers in (3.3) for the stationary and ergodic market. When applied to the conditionally log-optimal portfolio, we find [13]:

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \log(S_n^*) = W^* \right) = 1. \quad (3.13)$$

W^* is the optimal doubling rate given the infinite past.

$$W^* := \lim_{n \rightarrow \infty} \mathbb{E} \left[\max_{\mathbf{b} \in \mathcal{B}} \mathbb{E}[\log(\mathbf{b}^T \mathbf{X}_n) | \mathbf{X}_1, \dots, \mathbf{X}_{n-1}] \right] \quad (3.14)$$

4 Approximating Log-Optimal Investment

An analytic solution to the Kuhn-Tucker conditions, that describe the log-optimal portfolio, does not exist generally.

Cover [14] developed an iterative algorithm to approach the log-optimal portfolio which employed multiplicative steps motivated by the logarithmic utility.

Definition 4.0.1. (Algorithm for log investment [14]) Let $\mathbf{X}$ be a stock market vector distributed according to $F(\mathbf{x})$. Begin with a portfolio, $\mathbf{b}^0$, strictly inside the simplex. The n th iteration gives a portfolio vector with the following elements.

$$b_i^n = b_i^{n-1}(\nabla W(\mathbf{b}^{n-1}))_i = b_i^{n-1} \mathbb{E} \left[\frac{X_i}{(\mathbf{b}^{n-1})^T \mathbf{X}} \right] \quad (4.1)$$

In other words, each new iteration replaces the current portfolio with the portfolio expected after one period.

Proposition 4.0.1. Each iteration improves or does as well as the previous step in terms of the doubling rate it achieves. The increase in doubling rate due to one iteration is at least the Kullback-Leibler divergence between the new portfolio and the portfolio of the previous step.

$$W(\mathbf{b}^n) - W(\mathbf{b}^{n-1}) \geq D(\mathbf{b}^n \| \mathbf{b}^{n-1}) \geq 0 \quad (4.2)$$

Proof. Consider the following increase in doubling rate from the $(n-1)$ th iteration to the n th iteration.

$$\begin{aligned} W(\mathbf{b}^n) - W(\mathbf{b}^{n-1}) &= \mathbb{E}[\log((\mathbf{b}^n)^T \mathbf{X})] - \mathbb{E}[\log((\mathbf{b}^{n-1})^T \mathbf{X})] \\ &= \mathbb{E} \left[\log \left(\frac{(\mathbf{b}^n)^T \mathbf{X}}{(\mathbf{b}^{n-1})^T \mathbf{X}} \right) \right] \\ &= \mathbb{E} \left[\log \left(\sum_{i=1}^m \frac{b_i^n X_i}{(\mathbf{b}^{n-1})^T \mathbf{X}} \right) \right] \\ &= \mathbb{E} \left[\log \left(\sum_{i=1}^m \frac{b_i^{n-1} X_i}{(\mathbf{b}^{n-1})^T \mathbf{X}} \frac{b_i^n}{b_i^{n-1}} \right) \right] \end{aligned}$$

Trivially, $b_i^{n-1} X_i / (\mathbf{b}^{n-1})^T \mathbf{X}$ is non-negative and the sum over all $i \in \{1, \dots, m\}$ is equal to 1. As a result, the argument of the logarithm is a weighted average of b_i^n / b_i^{n-1} . Via concavity and Jensen's inequality, the logarithm of the weighted average is greater than or equal to the weighted average of the logarithms.

$$\begin{aligned} W(\mathbf{b}^n) - W(\mathbf{b}^{n-1}) &\geq \mathbb{E} \left[\sum_{i=1}^m \frac{b_i^{n-1} X_i}{(\mathbf{b}^{n-1})^T \mathbf{X}} \log \left(\frac{b_i^n}{b_i^{n-1}} \right) \right] \\ &= \sum_{i=1}^m b_i^{n-1} \mathbb{E} \left[\frac{X_i}{(\mathbf{b}^{n-1})^T \mathbf{X}} \right] \log \left(\frac{b_i^n}{b_i^{n-1}} \right) \\ &= \sum_{i=1}^m b_i^n \log \left(\frac{b_i^n}{b_i^{n-1}} \right) \\ &= D(\mathbf{b}^n \| \mathbf{b}^{n-1}) \\ &\geq 0 \end{aligned}$$

This final equality arises by definition of the Kullback-Leibler divergence. □

We shall not delve into the convergence of this algorithm. A rigorous examination is provided by Cover [14] who proves that the non-decreasing sequence of $W(\mathbf{b}^n)$ terminates at W^* by showing that the limit of $\mathbf{b}^n$ satisfies the Kuhn-Tucker conditions.

This iterative process (and all analysis as of yet) relies on the knowledge of the distribution of the stock market vectors, $F(\mathbf{x})$. In reality, human behaviour is erratic and the distribution is unknown and may not even exist. Theorem 4.0.1 highlights the dangers of attempting to guess this distribution.

Theorem 4.0.1. (Cost of approximation [8]) Let $\mathbf{X}$ be a stock market vector distributed according to $F(\mathbf{x})$ with density $f(\mathbf{x})$. Suppose an investor, perhaps in an attempt to determine the log-optimal portfolio, estimates the distribution as $\hat{F}(\mathbf{x})$ with density $\hat{f}(\mathbf{x})$. Let the actual log-optimal portfolio be $\mathbf{b}^*$ and the investor's estimate be $\hat{\mathbf{b}}^*$. The deviation from the optimal doubling rate is upper bounded by the Kullback-Leibler divergence between $f(\mathbf{x})$ and $\hat{f}(\mathbf{x})$.

$$W^* - W(\hat{\mathbf{b}}^*) \leq D(f \parallel \hat{f}) \quad (4.3)$$

Proof. Consider the deviation in the achieved doubling rate from the optimal doubling rate.

$$\begin{aligned} W^* - W(\hat{\mathbf{b}}^*) &= \mathbb{E}[\log(\mathbf{b}^{*T} \mathbf{X})] - \mathbb{E}[\log(\hat{\mathbf{b}}^{*T} \mathbf{X})] \\ &= \int \log(\mathbf{b}^{*T} \mathbf{x}) f(\mathbf{x}) d\mathbf{x} - \int \log(\hat{\mathbf{b}}^{*T} \mathbf{x}) f(\mathbf{x}) d\mathbf{x} \\ &= \int \log\left(\frac{\mathbf{b}^{*T} \mathbf{x}}{\hat{\mathbf{b}}^{*T} \mathbf{x}}\right) f(\mathbf{x}) d\mathbf{x} \\ &= \int \log\left(\frac{\mathbf{b}^{*T} \mathbf{x} \hat{f}(\mathbf{x})}{\hat{\mathbf{b}}^{*T} \mathbf{x} f(\mathbf{x})} \frac{f(\mathbf{x})}{\hat{f}(\mathbf{x})}\right) f(\mathbf{x}) d\mathbf{x} \\ &= \int \log\left(\frac{\mathbf{b}^{*T} \mathbf{x} \hat{f}(\mathbf{x})}{\hat{\mathbf{b}}^{*T} \mathbf{x} f(\mathbf{x})}\right) f(\mathbf{x}) d\mathbf{x} + \int \log\left(\frac{f(\mathbf{x})}{\hat{f}(\mathbf{x})}\right) f(\mathbf{x}) d\mathbf{x} \\ &= \int \log\left(\frac{\mathbf{b}^{*T} \mathbf{x} \hat{f}(\mathbf{x})}{\hat{\mathbf{b}}^{*T} \mathbf{x} f(\mathbf{x})}\right) f(\mathbf{x}) d\mathbf{x} + D(f \parallel \hat{f}) \end{aligned}$$

Applying Jensen's inequality to the first term, we have:

$$\begin{aligned} W^* - W(\hat{\mathbf{b}}^*) &\leq \log\left(\int \frac{\mathbf{b}^{*T} \mathbf{x} \hat{f}(\mathbf{x})}{\hat{\mathbf{b}}^{*T} \mathbf{x} f(\mathbf{x})} f(\mathbf{x}) d\mathbf{x}\right) + D(f \parallel \hat{f}) \\ &= \log\left(\int \frac{\mathbf{b}^{*T} \mathbf{x}}{\hat{\mathbf{b}}^{*T} \mathbf{x}} \hat{f}(\mathbf{x}) d\mathbf{x}\right) + D(f \parallel \hat{f}). \end{aligned}$$

$\hat{\mathbf{b}}^*$ is the log-optimal portfolio calculated assuming the stock market vectors are distributed with density $\hat{f}(\mathbf{x})$. Therefore, the following Kuhn-Tucker conditions are satisfied.

$$\mathbb{E}_{\hat{F}}\left[\frac{X_i}{\hat{\mathbf{b}}^{*T} \mathbf{X}}\right] = \int \frac{x_i}{\hat{\mathbf{b}}^{*T} \mathbf{x}} \hat{f}(\mathbf{x}) d\mathbf{x} \leq 1$$

A weighted average of this integral over all values of i would remain to be less than or equal to 1. Since $b_i^* \geq 0$ and $\sum_i b_i^* = 1$, multiplying our integral by b_i^* and summing over all i gives us the result.

$$\begin{aligned} \int \frac{\mathbf{b}^{*T} \mathbf{x}}{\hat{\mathbf{b}}^{*T} \mathbf{x}} \hat{f}(\mathbf{x}) d\mathbf{x} &\leq 1 \\ \log\left(\int \frac{\mathbf{b}^{*T} \mathbf{x}}{\hat{\mathbf{b}}^{*T} \mathbf{x}} \hat{f}(\mathbf{x}) d\mathbf{x}\right) &\leq 0 \\ W^* - W(\hat{\mathbf{b}}^*) &\leq D(f \parallel \hat{f}) \end{aligned}$$

□

Consider a black swan event that our models suggest shouldn't happen, $\hat{f}(\mathbf{x}) = 0$, but can occur in reality, $f(\mathbf{x}) > 0$. Then the divergence reaches infinity and the investor's doubling rate can stray significantly from the optimal doubling rate.

In other words, predicting the stock market vector distribution is problematic and should be avoided. This warrants the following definition.

Definition 4.0.2. (*Universal portfolio [15]*) Let $\mathbf{x}_1, \mathbf{x}_2, \dots$ be arbitrary stock market vectors which need not follow a distribution. A universal portfolio is a causal investment strategy, $\{\mathbf{b}_i\}_{i \in \mathbb{N}}$, that performs asymptotically as well as the best constant rebalanced portfolio in hindsight with knowledge of the realised stock market vectors.

Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be the observed stock market vectors from n periods. Then the best constant rebalanced portfolio, $\mathbf{b}_n^B$ ², chosen with hindsight is defined by:

$$\mathbf{b}_n^B := \arg \max_{\mathbf{b} \in \mathcal{B}} \prod_{i=1}^n \mathbf{b}^T \mathbf{x}_i. \quad (4.4)$$

The log-optimal portfolio can't be used as a benchmark because the expectation in (3.5) is undefined when there is no underlying distribution for the stock market vectors.

Remark 4.0.1. The rationale behind the use of $\mathbf{b}_n^B$ as the benchmark portfolio is clear when the market is i.i.d. [25]. Since the logarithm is strictly concave and the stock market vectors are not random, $\mathbf{b}_n^B$ may be written as:

$$\mathbf{b}_n^B = \arg \max_{\mathbf{b} \in \mathcal{B}} \sum_{i=1}^n \log(\mathbf{b}^T \mathbf{x}_i) = \arg \max_{\mathbf{b} \in \mathcal{B}} \frac{1}{n} \sum_{i=1}^n \log(\mathbf{b}^T \mathbf{x}_i). \quad (4.5)$$

Therefore, via the strong law of large numbers in (3.3), $\mathbf{b}_n^B$ converges almost surely to the log-optimal portfolio, $\mathbf{b}^*$, as $n \rightarrow \infty$. So a universal portfolio in an i.i.d. market must perform asymptotically as well as the log-optimal portfolio.

Two such universal portfolios are discussed below.

4.1 The Empirical Bayes Portfolio

Cover and Gluss [16] addressed this problem using Robbins' [17] empirical Bayes method. The distribution of stock market vectors is estimated from the vectors observed up to that time.

Definition 4.1.1. (*Empirical distribution*) Say an investor has witnessed the stock market vectors, $\mathbf{x}_1, \dots, \mathbf{x}_n$, after n time periods. The empirical distribution of the stock market vectors, $F_n(\mathbf{x})$, gives each observed $\mathbf{x}_i$ an equal weighting. The empirical density, $f_n(\mathbf{x})$, is defined by:

$$f_n(\mathbf{x}) := \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\mathbf{x}_i = \mathbf{x}\}. \quad (4.6)$$

Instead of using the empirical distribution for a plug-in estimator of the log-optimal portfolio, Cover and Gluss [16] use an alternative distribution that arises from the use of Blackwell's approach-exclusion theorem [18].

²The reader may understand this B notation as reflecting that this constant rebalanced portfolio is 'best' or reflecting the fact it acts as the 'benchmark' for universal portfolios.

Assume that the stock market vectors are all from a finite set, χ , with cardinality M .

$$\chi := \{\mathbf{a}_1, \dots, \mathbf{a}_M\} \quad (4.7)$$

Let $\mathbf{p} \in \mathbb{R}^M$ be a probability vector where each element represents the probability of attaining each of the M possible stock market vectors in χ . The space of probability vectors is a probability simplex, $\mathcal{P}$, of dimension $M - 1$. The empirical probability vector after n periods can be calculated using (4.6).

$$\mathbf{p}_n = (f_n(\mathbf{a}_1), \dots, f_n(\mathbf{a}_M)) \quad (4.8)$$

Definition 4.1.2. (*Bayes envelope [16]*) The Bayes envelope, $\mathcal{E}$, is the convex set consisting of all pairs of a probability vector, $\mathbf{p}$, and a doubling rate, W , such that the doubling rate is at least the doubling rate obtained from the log-optimal portfolio under the assumption the stock market vectors are distributed according to $\mathbf{p}$.

$$\mathcal{E} := \{(\mathbf{p}, W) : \mathbf{p} \in \mathcal{P}, W \geq W^*(\mathbf{p})\} \subset \mathbb{R}^{M+1} \quad (4.9)$$

Note that the doubling rate will now optionally take the distribution as an argument for cases where it is not clear.

Definition 4.1.3. (*Empirical Bayes portfolio [16]*) Let $\mathbf{x}_1, \mathbf{x}_2, \dots$ be a sequence of stock market vectors from the finite set χ . After n periods, an investor with a causal portfolio will have an empirical distribution, $\mathbf{p}_n$, from the stock market vectors they have observed and a realised doubling rate, $W_n := \frac{1}{n} \log(S_n)$. Let $(\hat{\mathbf{p}}_n, \hat{W}_n) \in \mathcal{E}$ be the closest pair in the Bayes envelope to the investor's $(\mathbf{p}_n, W_n)$. The empirical Bayes portfolio is the following causal investment strategy.

$$\mathbf{b}_{n+1}^{EB} := \mathbf{b}^*(\hat{\mathbf{p}}_n) \quad (4.10)$$

In other words, each updated portfolio is the log-optimal portfolio under the assumption that the stock market vectors are distributed according to the current $\hat{\mathbf{p}}$.

Theorem 4.1.1. (*Asymptotic optimality of the empirical Bayes portfolio [16]*) Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be a sequence of stock market vectors from the finite set χ with cardinality M . Suppose that all possible stock market vectors, $\{\mathbf{a}_i\}$ for $i \in \{1, \dots, M\}$, are bounded such that $|\log(a_{ij})| \leq L$. Then the realised doubling rate, achieved using the empirical Bayes portfolio, converges asymptotically to the optimal doubling rate in hindsight.

$$\frac{1}{n} \log(S_n^{EB}) \geq W^*(\mathbf{p}_n) - \frac{(L\sqrt{M} + 1)\sqrt{4L^2 + 2}}{\sqrt{n}} \quad (4.11)$$

Proof. The assumption that $|\log(a_{ij})| \leq L$ means the realised doubling rates are similarly bounded, $|W| \leq L$. Then the space of all pairs of probability vectors and doubling rates is the Cartesian product, $\mathcal{P} \times [-L, L]$.

$$\mathcal{U} := \{(\mathbf{p}, W) : \mathbf{p} \in \mathcal{P}, |W| \leq L\} \subset \mathbb{R}^{M+1}$$

$\mathcal{U}$ is trivially bounded. The Bayes envelope, $\mathcal{E}$, is a subset of $\mathcal{U}$ which we shall take to be closed and convex without proof. See Cover and Gluss [16] for a more comprehensive treatment.

Let $\mathbf{q}_i = (\mathbf{p}_i, W_i)$ be a sequence of points in $\mathcal{U}$ with $i \in \{1, \dots, n\}$. Here, $\mathbf{p}_i$ is the empirical probability vector based on i observed stock market vectors and W_i is the average logarithmic return after i periods. $\mathbf{q}_i$ represents a rolling average which shall be guided towards the Bayes envelope.

Say $\mathbf{q}_i$ is the rolling average of points $\mathbf{y}_i \in \mathbb{R}^{M+1}$.

$$\mathbf{q}_k = \frac{1}{k} \sum_{i=1}^k \mathbf{y}_i$$

Working backwards, we find that $\mathbf{y}_i$ must be defined as:

$$\mathbf{y}_i = (\mathbf{I}(\mathbf{x}_i), \log(\mathbf{b}_i^T \mathbf{x}_i))$$

where $\mathbf{I}(\mathbf{x}_i) \in \mathbb{R}^M$ is the unit vector with a 1 in the j th coordinate where $\mathbf{a}_j = \mathbf{x}_i$. A quick check shows this yields the rolling average as required.

$$\begin{aligned} \frac{1}{k} \sum_{i=1}^k \mathbf{y}_i &= \frac{1}{k} \sum_{i=1}^k (\mathbf{I}(\mathbf{x}_i), \log(\mathbf{b}_i^T \mathbf{x}_i)) \\ &= \left(\frac{1}{k} \sum_{i=1}^k \mathbf{I}(\mathbf{x}_i), \frac{1}{k} \log(S_k) \right) \\ &= (\mathbf{p}_k, W_k) \\ &= \mathbf{q}_k \end{aligned}$$

When $\mathbf{q}_k$ has not yet reached the Bayes envelope, $\mathcal{E}$, there is a unique point, $\mathbf{t}_k = (\hat{\mathbf{p}}_k, \hat{W}_k) \in \mathbb{R}^{M+1}$, in $\mathcal{E}$ which is closest (in Euclidean norm) to $\mathbf{q}_k$. Uniqueness is guaranteed by the convexity of $\mathcal{E}$.

Define a supporting hyperplane, $\mathcal{H}_k$, from which the point $\mathbf{y}_{k+1}$ will be drawn to find the latest rolling average, $\mathbf{q}_{k+1}$. This supporting hyperplane should contain $\mathbf{t}_k$ and be perpendicular to $\mathbf{t}_k - \mathbf{q}_k$.

As $\mathbf{t}_k$ lies on the boundary of the Bayes envelope, $\hat{W}_k = W^*(\hat{\mathbf{p}}_k)$. Select the empirical Bayes portfolio, $\mathbf{b}_{k+1}^{EB} = \mathbf{b}^*(\hat{\mathbf{p}}_k)$, and consider the following hyperplane.

$$\mathcal{H}_k = \{(\mathbf{p}, W(\mathbf{b}^*(\hat{\mathbf{p}}_k), \mathbf{p})) : \mathbf{p} \in \mathcal{P}\}$$

The doubling rate is maximised with the empirical Bayes portfolio when the distribution is $\hat{\mathbf{p}}_k$. For all probability vectors, $W^*(\mathbf{p}) \geq W(\mathbf{b}^*(\hat{\mathbf{p}}_k), \mathbf{p})$. We have equality at $\hat{\mathbf{p}}_k$ and so this is a supporting hyperplane to the Bayes envelope at $\mathbf{t}_k$.

Furthermore, $\mathbf{y}_{k+1}$ lies on this supporting hyperplane, $\mathcal{H}_k$, as required. This is evident when $\mathbf{y}_{k+1}$ is written as:

$$\mathbf{y}_{k+1} = (\mathbf{I}(\mathbf{x}_{k+1}), \log(\mathbf{b}_k^{*T}(\hat{\mathbf{p}}_k) \mathbf{x}_{k+1})) = (\mathbf{I}(\mathbf{x}_{k+1}), W(\mathbf{b}^*(\hat{\mathbf{p}}_k), \mathbf{I}(\mathbf{x}_{k+1})))$$

where we have identified $\mathbf{p}$ as $\mathbf{I}(\mathbf{x}_{k+1})$.

This analysis allows us to invoke Blackwell's approach-exclusion theorem as adopted by Cover and Gluss [16]. This theorem roughly states that, using the prescribed empirical Bayes portfolio, the rolling average, $\mathbf{q}_n$, approaches the Bayes envelope (our target set) as $n \rightarrow \infty$. More specifically,

$$\|\mathbf{q}_n - \mathbf{t}_n\|_2 \leq \frac{D}{\sqrt{n}}$$

where D is the diameter of $\mathcal{U}$; the largest possible distance between two elements in $\mathcal{U}$.

$$\begin{aligned}
D^2 &= \max_{\mathbf{p}, \mathbf{q} \in \mathcal{P}} \|\mathbf{p} - \mathbf{q}\|_2^2 + \max_{W, V \in [-L, L]} \|W - V\|_2^2 \\
&= \left\| \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix} \right\|_2^2 + \|L - (-L)\|_2^2 \\
&= 2 + 4L^2
\end{aligned}$$

The probability vector is a component of $\mathbf{q}_n$ and $\mathbf{t}_n$ so the following weaker inequality is also true.

$$\|\mathbf{p}_n - \hat{\mathbf{p}}_n\|_2 \leq \sqrt{\frac{2 + 4L^2}{n}}$$

The left-hand side of this inequality may be converted to a 1-norm using the Cauchy-Schwarz inequality.

$$\begin{aligned}
\|\mathbf{p}_n - \hat{\mathbf{p}}_n\|_1 &= \mathbf{1}_M^T \begin{pmatrix} |p_{n1} - \hat{p}_{n1}| \\ |p_{n2} - \hat{p}_{n2}| \\ \vdots \\ |p_{nM} - \hat{p}_{nM}| \end{pmatrix} \\
&\leq \|\mathbf{1}_M\|_2 \|\mathbf{p}_n - \hat{\mathbf{p}}_n\|_2 \\
&= \sqrt{M} \|\mathbf{p}_n - \hat{\mathbf{p}}_n\|_2 \\
&\leq \sqrt{\frac{M(2 + 4L^2)}{n}}
\end{aligned}$$

This expression may be written in terms of doubling rates using the fact that $W^*(\mathbf{p})$ is Lipschitz continuous with respect to $\mathbf{p}$. This was proved by Cover and Gluss [16] and arises from the stock market vectors being bounded.

$$\begin{aligned}
|W^*(\mathbf{p}_n) - W^*(\hat{\mathbf{p}}_n)| &\leq L \|\mathbf{p}_n - \hat{\mathbf{p}}_n\|_1 \\
&\leq L \sqrt{\frac{M(2 + 4L^2)}{n}}
\end{aligned}$$

The doubling rates are also a component of $\mathbf{q}_n$ and $\mathbf{t}_n$ so we have another weaker inequality from the main result of Blackwell's theorem.

$$|W_n - W^*(\hat{\mathbf{p}}_n)| \leq \frac{D}{\sqrt{n}}$$

Coupling this inequality with the triangle inequality, we arrive at the desired result.

$$\begin{aligned}
|W_n - W^*(\mathbf{p}_n)| &= |W_n - W^*(\hat{\mathbf{p}}_n) + W^*(\hat{\mathbf{p}}_n) - W^*(\mathbf{p}_n)| \\
&\leq |W_n - W^*(\hat{\mathbf{p}}_n)| + |W^*(\mathbf{p}_n) - W^*(\hat{\mathbf{p}}_n)| \\
&\leq \sqrt{\frac{2 + 4L^2}{n}} + L \sqrt{\frac{M(2 + 4L^2)}{n}} \\
&= \frac{(L\sqrt{M} + 1)\sqrt{4L^2 + 2}}{\sqrt{n}}
\end{aligned}$$

This absolute value implies two inequalities.

$$W_n \geq W^*(\mathbf{p}_n) - \frac{(L\sqrt{M} + 1)\sqrt{4L^2 + 2}}{\sqrt{n}}$$

$$\frac{1}{n} \log(S_n^{EB}) \geq W^*(\mathbf{p}_n) - \frac{(L\sqrt{M} + 1)\sqrt{4L^2 + 2}}{\sqrt{n}}$$

□

The best constant rebalanced portfolio is the log-optimal portfolio under the empirical distribution. Therefore, $W^*(\mathbf{p}_n)$ is the doubling rate achieved using the best constant rebalanced portfolio in hindsight. Theorem 4.1.1 proves that use of the causal empirical Bayes portfolio allows us to perform as well as a clairvoyant investor, with knowledge of the empirical distribution of stock market vectors, in the long run.

4.2 The Universal Portfolio

Cover [15] continued to better the field of universal portfolios by eliminating the need for the stock market vectors to come from a finite set.

Definition 4.2.1. (*The Universal portfolio*³[15]) Let $\mathbf{x}_1, \mathbf{x}_2, \dots$ be an arbitrary sequence of stock market vectors. The Universal portfolio is the causal investment strategy that begins with an equally-weighted portfolio, $\mathbf{b}_1^U = \frac{1}{m}\mathbf{1}_m$, and adapts these weights via:

$$\mathbf{b}_{k+1}^U := \frac{\int_{\mathcal{B}} \mathbf{b} S_k(\mathbf{b}) d\mathbf{b}}{\int_{\mathcal{B}} S_k(\mathbf{b}) d\mathbf{b}}. \quad (4.12)$$

The Universal portfolio at period $k + 1$ is the average of all portfolios in the simplex weighted by the wealth relative accumulated up until period k .

Similarly, the wealth relative after n periods using the Universal portfolio is the average wealth of all portfolios in the simplex.

$$S_n^U = \frac{\int_{\mathcal{B}} S_n(\mathbf{b}) d\mathbf{b}}{\int_{\mathcal{B}} d\mathbf{b}} \quad (4.13)$$

Previously, asymptotic optimality of a portfolio has been expressed as the convergence of the realised doubling rate towards the optimal doubling rate with time. We shall prove a stronger result comparing the wealth of the Universal portfolio to that of the best constant rebalanced portfolio. For such an analysis, the curvature of $S_n(\mathbf{b})$ must be considered to determine how close the Universal portfolio wealth can get to the maximum.

Definition 4.2.2. (*Sensitivity matrix function* [15]) The sensitivity matrix function, $J(\mathbf{b})$, is a measure of the curvature of the doubling rate and is defined as the negative of the Hessian matrix of $W(\mathbf{b})$, $\mathbf{H}_W$. Let the distribution of stock market vectors associated with the definition of $W(\mathbf{b})$ be $F(\mathbf{x})$. Explicitly, the elements of this square matrix are given by:

$$J_{ij}(\mathbf{b}) := -\frac{\partial^2 W(b_1, \dots, b_{m-1}, 1 - \sum_{k=1}^{m-1} b_k)}{\partial b_i \partial b_j} = \int \frac{(x_i - x_m)(x_j - x_m)}{(\mathbf{b}^T \mathbf{x})^2} dF(\mathbf{x}) \quad (4.14)$$

where $i, j \in \{1, \dots, m - 1\}$. This separation of the m th asset alludes to the use of alternative portfolio coordinates (as seen in the proof of Theorem 4.2.1) that remove the m th element of the portfolio vector which is made redundant by the constraint $\sum_i b_i = 1$.

³To remove ambiguity between Cover's 1991 'Universal Portfolios' algorithm and the general universal portfolios, of which the empirical Bayes portfolio is an example, we shall call this specific portfolio the Universal portfolio (singular and capitalised).

This may seem like a step backwards with the assumption of a distribution but this definition shall only be used with empirical distributions, $F_n(\mathbf{x})$. When this is the case, the matrix will be written as $J_n(\mathbf{b})$.

Theorem 4.2.1. (*Asymptotic wealth of the Universal portfolio [15]*) Consider an arbitrary sequence of stock market vectors, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, containing finite, non-zero price relatives, $x \in [a, d]$. As this sequence is arbitrary, there is no guarantee that the realised doubling rate, curvature or best constant rebalanced portfolio settle down to a constant. Therefore, we consider a subsequence of periods, $n_1, n_2, \dots$, such that $W_n(\mathbf{b}) \nearrow W(\mathbf{b})$, $\mathbf{b}_n^B \rightarrow \mathbf{b}^B$ and $J_n(\mathbf{b}_n^B) \rightarrow J^B$ ⁴. This limit of the realised doubling rate, $W(\mathbf{b})$, shall be assumed to be strictly concave with bounded third partial derivatives and a maximum attained at $\mathbf{b}^B$ within the interior of the simplex.

Under these assumptions, the ratio of the wealth of the Universal portfolio to that of the best constant rebalanced portfolio has the following asymptotic behaviour along the subsequence.

$$\frac{S_n^U}{S_n^B} \sim \left(\frac{2\pi}{n} \right)^{\frac{m-1}{2}} \frac{(m-1)!}{|J^B|^{1/2}} \quad (4.15)$$

Proof. The portfolio simplex, $\mathcal{B}$, is $(m-1)$ -dimensional but there are m elements of the portfolio vector. It is beneficial to remove the m th coordinate by defining it with the constraint $\sum_i b_i = 1$.

$$\mathcal{C} := \left\{ (c_1, \dots, c_{m-1}) : c_j \geq 0, \sum_{i=1}^{m-1} c_i \leq 1 \right\}$$

A portfolio, $\mathbf{b} \in \mathcal{B}$, is related to $\mathbf{c} \in \mathcal{C}$ via:

$$\mathbf{b}(\mathbf{c}) = \left(c_1, \dots, c_{m-1}, 1 - \sum_{i=1}^{m-1} c_i \right).$$

Let $\mathbf{c}^B$ be the vector corresponding to the best constant rebalanced portfolio, $\mathbf{b}^B$. To achieve a lower bound on the wealth of a portfolio after n days, $S_n(\mathbf{c})$, we can use Taylor's theorem to expand $W_n(\mathbf{c})$ about $\mathbf{c}^B$.

$$W_n(\mathbf{c}) = W_n(\mathbf{c}^B) + \nabla W_n(\mathbf{c}^B)^T (\mathbf{c} - \mathbf{c}^B) + \frac{1}{2} (\mathbf{c} - \mathbf{c}^B)^T \mathbf{H}_{W_n}(\mathbf{c}^B) (\mathbf{c} - \mathbf{c}^B) + R_3(\mathbf{c})$$

As $\mathbf{c}^B$ is the maximiser of $W_n(\mathbf{c})$, the first-order term vanishes. Substituting in for the third-order remainder term, R_3 , and the sensitivity matrix $J_n^B := J_n(\mathbf{b}_n^B)$, the expansion becomes:

$$W_n(\mathbf{c}) = W_n(\mathbf{c}^B) - \frac{1}{2} (\mathbf{c} - \mathbf{c}^B)^T J_n^B (\mathbf{c} - \mathbf{c}^B) + \frac{1}{6} \sum_{i=1}^{m-1} \sum_{j=1}^{m-1} \sum_{k=1}^{m-1} \frac{\partial^3 W_n(\tilde{\mathbf{c}})}{\partial c_i \partial c_j \partial c_k} (c_i - c_i^B)(c_j - c_j^B)(c_k - c_k^B)$$

where $\tilde{\mathbf{c}}$ is a weighted average of $\mathbf{c}$ and $\mathbf{c}^B$. For some $\lambda \in [0, 1]$,

$$\tilde{\mathbf{c}} = \lambda \mathbf{c}^B + (1 - \lambda) \mathbf{c}.$$

⁴We stray slightly from the notation of Cover [15] with respect to the best constant rebalanced portfolio. Cover labels all related variables with a *. This notation, however, is used here to represent variables relating to the log-optimal portfolio. Now that we are working with no assumed distribution, we employ the ^B notation to differentiate between the log-optimal framework using an estimated or assumed distribution and that using the empirical distribution.

Consider the partial derivatives of $W_n(\mathbf{c})$. Treat the arbitrary stock market vectors as realised values of random vectors distributed according to the empirical distribution, $F_n(\mathbf{x})$.

$$\begin{aligned} W_n(\mathbf{c}) &= \mathbb{E}_{F_n}[\log(S(\mathbf{c}))] = \mathbb{E}_{F_n} \left[\log \left(\sum_{i=1}^{m-1} c_i X_i + \left(1 - \sum_{i=1}^{m-1} c_i \right) X_m \right) \right] \\ \frac{\partial W_n(\mathbf{c})}{\partial c_k} &= \mathbb{E}_{F_n} \left[\frac{X_k - X_m}{S(\mathbf{c})} \right] \\ \frac{\partial^2 W_n(\mathbf{c})}{\partial c_j \partial c_k} &= \mathbb{E}_{F_n} \left[-\frac{(X_j - X_m)(X_k - X_m)}{S^2(\mathbf{c})} \right] \\ \frac{\partial^3 W_n(\mathbf{c})}{\partial c_i \partial c_j \partial c_k} &= \mathbb{E}_{F_n} \left[\frac{2(X_i - X_m)(X_j - X_m)(X_k - X_m)}{S^3(\mathbf{c})} \right] \end{aligned}$$

Substitute this into the expansion of $W_n(\mathbf{c})$.

$$W_n(\mathbf{c}) = W_n(\mathbf{c}^B) - \frac{1}{2}(\mathbf{c} - \mathbf{c}^B)^T J_n^B(\mathbf{c} - \mathbf{c}^B) + \frac{1}{3} \sum_{i,j,k=1}^{m-1} (c_i - c_i^B)(c_j - c_j^B)(c_k - c_k^B) \mathbb{E}_{F_n} \left[\frac{(X_i - X_m)(X_j - X_m)(X_k - X_m)}{S^3(\tilde{\mathbf{c}})} \right]$$

This realised doubling rate shall first be bounded from below through the claim that the remainder term must be at least the negative of its absolute magnitude.

$$W_n(\mathbf{c}) \geq W_n(\mathbf{c}^B) - \frac{1}{2}(\mathbf{c} - \mathbf{c}^B)^T J_n^B(\mathbf{c} - \mathbf{c}^B) - |R_3(\mathbf{c})|$$

$$|R_3(\mathbf{c})| \leq \frac{1}{3} \sum_{i,j,k=1}^{m-1} |c_i - c_i^B| |c_j - c_j^B| |c_k - c_k^B| \mathbb{E}_{F_n} \left[\frac{|X_i - X_m| |X_j - X_m| |X_k - X_m|}{|S^3(\tilde{\mathbf{c}})|} \right]$$

The expectation term may be bounded through the condition on the price relatives $a \leq X_i \leq d$.

$$|X_i - X_m| \leq 2d$$

Although we could construct a tighter bound here, this is all that is required.

$$S(\tilde{\mathbf{c}}) = \sum_{i=1}^m \tilde{b}_i X_i \geq a \sum_{i=1}^m \tilde{b}_i = a$$

Substituting these inequalities into the remainder term, we find:

$$\begin{aligned} |R_3(\mathbf{c})| &\leq \frac{8d^3}{3a^3} \sum_{i,j,k=1}^{m-1} |c_i - c_i^B| |c_j - c_j^B| |c_k - c_k^B| \\ &= \frac{8d^3}{3a^3} \left(\sum_{i=1}^{m-1} |c_i - c_i^B| \right)^3. \end{aligned}$$

Via the Cauchy-Schwarz inequality,

$$\begin{aligned} |R_3(\mathbf{c})| &\leq \frac{8d^3}{3a^3} (\sqrt{m-1} \|\mathbf{c} - \mathbf{c}^B\|_2)^3 \\ &\leq \frac{8d^3}{3a^3} m^{3/2} \|\mathbf{c} - \mathbf{c}^B\|_2^3. \end{aligned}$$

The lower bound on the realised doubling rate translates to a lower bound on the wealth.

$$\begin{aligned}\frac{1}{n} \log(S_n(\mathbf{c})) &\geq W_n(\mathbf{c}^B) - \frac{1}{2}(\mathbf{c} - \mathbf{c}^B)^T J_n^B(\mathbf{c} - \mathbf{c}^B) - \frac{8d^3}{3a^3} m^{3/2} \|\mathbf{c} - \mathbf{c}^B\|_2^3 \\ S_n(\mathbf{c}) &\geq \exp\left(nW_n(\mathbf{c}^B) - \frac{n}{2}(\mathbf{c} - \mathbf{c}^B)^T J_n^B(\mathbf{c} - \mathbf{c}^B) - \frac{8d^3}{3a^3} nm^{3/2} \|\mathbf{c} - \mathbf{c}^B\|_2^3\right) \\ &= S_n^B \exp\left(-\frac{n}{2}(\mathbf{c} - \mathbf{c}^B)^T J_n^B(\mathbf{c} - \mathbf{c}^B) - \frac{8d^3}{3a^3} nm^{3/2} \|\mathbf{c} - \mathbf{c}^B\|_2^3\right)\end{aligned}$$

Now consider the wealth of the Universal portfolio given in (4.13).

$$S_n^U = \frac{\int_{\mathcal{C}} S_n(\mathbf{c}) d\mathbf{c}}{\int_{\mathcal{C}} d\mathbf{c}} = (m-1)! \int_{\mathcal{C}} S_n(\mathbf{c}) d\mathbf{c}$$

The second equality comes from the standard result for the volume of a probability simplex.

$$\frac{S_n^U}{S_n^B} \geq (m-1)! \int_{\mathcal{C}} \exp\left(-\frac{n}{2}(\mathbf{c} - \mathbf{c}^B)^T J_n^B(\mathbf{c} - \mathbf{c}^B) - \frac{8d^3}{3a^3} nm^{3/2} \|\mathbf{c} - \mathbf{c}^B\|_2^3\right) d\mathbf{c}$$

Let $\mathbf{u} = \sqrt{n}(\mathbf{c} - \mathbf{c}^B) \in \mathbb{R}^{m-1}$.

$$\frac{S_n^U}{S_n^B} \geq (m-1)! \int_{\mathcal{U}} \exp\left(-\frac{1}{2} \mathbf{u}^T J_n^B \mathbf{u} - \frac{8d^3}{3a^3} \frac{m^{3/2}}{\sqrt{n}} \|\mathbf{u}\|_2^3\right) \left(\frac{1}{\sqrt{n}}\right)^{m-1} d\mathbf{u}$$

The integration is now over $\mathcal{U}$, the space of all possible vectors, $\mathbf{u}$, generated from $\mathcal{C}$. This space depends on n and we cannot simply consider the limit of this integral yet. Let $\mathcal{B}_R$ be a ball of fixed radius R . For sufficiently large n , $\mathcal{B}_R \subset \mathcal{U}$ and the following is true.

$$\int_{\mathcal{U}} \exp\left(-\frac{1}{2} \mathbf{u}^T J_n^B \mathbf{u} - \frac{8d^3}{3a^3} \frac{m^{3/2}}{\sqrt{n}} \|\mathbf{u}\|_2^3\right) d\mathbf{u} \geq \int_{\mathcal{B}_R} \exp\left(-\frac{1}{2} \mathbf{u}^T J_n^B \mathbf{u} - \frac{8d^3}{3a^3} \frac{m^{3/2}}{\sqrt{n}} \|\mathbf{u}\|_2^3\right) d\mathbf{u}$$

We can now calculate the asymptotic lower bound of this wealth ratio.

$$\liminf_{n \rightarrow \infty} \frac{S_n^U n^{\frac{m-1}{2}}}{S_n^B (m-1)!} \geq \int_{\mathcal{B}_R} \exp\left(-\frac{1}{2} \mathbf{u}^T J^B \mathbf{u}\right) d\mathbf{u}$$

As $n \rightarrow \infty$, R may also be taken to infinity, allowing the Gaussian integral to be evaluated.

$$\begin{aligned}\liminf_{n \rightarrow \infty} \frac{S_n^U n^{\frac{m-1}{2}}}{S_n^B (m-1)!} &\geq \frac{(2\pi)^{\frac{m-1}{2}}}{|J^B|^{1/2}} \\ \liminf_{n \rightarrow \infty} \frac{S_n^U}{S_n^B} &\geq \left(\frac{2\pi}{n}\right)^{\frac{m-1}{2}} \frac{(m-1)!}{|J^B|^{1/2}}\end{aligned}$$

Half the proof is now complete. We must now show that the asymptotic upper bound is equal to the asymptotic lower bound just determined.

$$\begin{aligned}S_n^U &= (m-1)! \int_{\mathcal{C}} S_n(\mathbf{c}) d\mathbf{c} \\ &= (m-1)! \int_{\mathcal{C}} \exp(nW_n(\mathbf{c})) d\mathbf{c}\end{aligned}$$

The subsequence of time periods, along which we prove this asymptotic behaviour, is defined to have the realised doubling rate increase up to the limit of $W(\mathbf{b})$. As the exponential function is increasing, this becomes an upper bound on the wealth of the Universal portfolio.

$$S_n^U \leq (m-1)! \int_{\mathcal{C}} \exp(nW(\mathbf{c})) d\mathbf{c}$$

$W(\mathbf{b})$ is uniquely maximised by the portfolio $\mathbf{b}^B$ in the interior of the simplex. Furthermore, $W(\mathbf{b})$ is strictly concave with bounded third partial derivatives. These assumptions allow the use of Laplace's method of integration here. In a similar fashion that achieved the lower bound, this method approximates $W(\mathbf{c})$ about the maximum because these points will be most significant as $n \rightarrow \infty$.

$$\begin{aligned} \int_{\mathcal{C}} \exp(nW(\mathbf{c})) d\mathbf{c} &\sim \exp(nW(\mathbf{c}^B)) \left(\frac{2\pi}{n} \right)^{\frac{m-1}{2}} \frac{1}{|\mathbf{H}_W(\mathbf{c}^B)|^{1/2}} \\ &\sim S_n^B \left(\frac{2\pi}{n} \right)^{\frac{m-1}{2}} \frac{1}{|J^B|^{1/2}} \end{aligned}$$

This leads to the following asymptotic upper bound for the ratio of the wealth of the Universal portfolio and the best constant rebalanced portfolio.

$$\limsup_{n \rightarrow \infty} \frac{S_n^U}{S_n^B} \leq \left(\frac{2\pi}{n} \right)^{\frac{m-1}{2}} \frac{(m-1)!}{|J^B|^{1/2}}$$

As $n \rightarrow \infty$, the sequence of wealth ratios is bounded above and below by the same value which it must, therefore, converge to.

$$\frac{S_n^U}{S_n^B} \sim \left(\frac{2\pi}{n} \right)^{\frac{m-1}{2}} \frac{(m-1)!}{|J^B|^{1/2}}$$

□

Proposition 4.2.1. *In the limit as $n \rightarrow \infty$, the Universal portfolio achieves the same doubling rate of wealth as the best constant rebalanced portfolio chosen with hindsight.*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n^U}{S_n^B} \right) = 0 \quad (4.16)$$

Proof. Follows directly from Theorem 4.2.1.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{S_n^U}{S_n^B} \right) &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left(\frac{2\pi}{n} \right)^{\frac{m-1}{2}} \frac{(m-1)!}{|J^B|^{1/2}} \right) \\ &= -\frac{m-1}{2} \lim_{n \rightarrow \infty} \frac{\log(n)}{n} \\ &= 0 \end{aligned}$$

□

When we remove the assumption that the best constant rebalanced portfolio lies in the interior of the simplex (required for the use of Laplace's method), Theorem 4.2.1 remains to be true and the dimension, currently $m-1$, is reduced.

Due to the commutativity of multiplication, the wealth of the Universal portfolio, S_n^U , is invariant under permutations of the stock market vector sequence, $\mathbf{x}_1, \dots, \mathbf{x}_n$ [15]. This property demonstrates the effect of crashes in the stock market on the Universal portfolio. Consider a period of extremely poor returns from k to k' . Then this gives a wealth identical to if $\mathbf{x}_k, \dots, \mathbf{x}_{k'}$ had been evenly distributed among all n periods, which is reminiscent of typical investing.

5 Implementing the Universal Portfolio

Real-world usage of the Universal portfolio cannot rely on the averaging over all possible portfolios in the simplex.

Definition 5.0.1. (*Monte Carlo Universal portfolio*) Imagine there are N portfolio managers each with a constant rebalanced portfolio, $\mathbf{b}^{(i)}$, and a wealth relative, $S_n^{(i)}$, after n periods. Let $\mathbf{b}^{(1)}, \dots, \mathbf{b}^{(N)}$ be i.i.d. random variables drawn uniformly from the portfolio simplex. A Monte Carlo approximation to the Universal portfolio distributes wealth amongst these managers dependent on their past performance.

$$\hat{\mathbf{b}}_{k+1}^U := \frac{\sum_{i=1}^N S_k^{(i)} \mathbf{b}^{(i)}}{\sum_{i=1}^N S_k^{(i)}} \quad (5.1)$$

$$\hat{S}_k^U := \frac{1}{N} \sum_{i=1}^N S_n^{(i)} \quad (5.2)$$

By the strong law of large numbers, this approximation converges almost surely to the Universal portfolio as $N \rightarrow \infty$. The more portfolio managers in our Monte Carlo method, N , and the longer time period we invest over, n , the closer we approximate the best constant rebalanced portfolio in hindsight.

Another unrealistic assumption we have adopted is that of a frictionless market without transaction costs. This simplification allowed the daily rebalancing of our portfolios without being charged commission or experiencing a bid-ask spread.

Definition 5.0.2. (*Rebalancing transaction costs*) Say we hold a portfolio, $\mathbf{b}_k$, at time k . This portfolio witnesses a stock market vector, $\mathbf{X}_k$, which alters the proportions to give $\mathbf{b}'_k$. If we have calculated the optimal portfolio for the next period to be $\mathbf{b}_{k+1}$ then rebalancing is required to bring us from $\mathbf{b}'_k$ to $\mathbf{b}_{k+1}$. In our simple model inspired by Blum and Kalai [19], we assume the transaction costs, T , are proportional to the value of assets bought and sold during rebalancing.

$$T := c S_k \|\mathbf{b}'_k - \mathbf{b}_{k+1}\|_1 \quad (5.3)$$

The constant of proportionality, c , should be chosen to represent the magnitude of transaction costs typical of the market. Mathematically, we should focus on $0 \leq c \leq 0.5$ to ensure transaction costs cannot exceed total wealth. In real financial markets, c is very unlikely to exceed even 0.1. Larger c could reflect brokers with higher commissions or larger bid-ask spreads due to illiquid markets.

The introduction of transaction costs dissuades investors from rebalancing portfolios. With large transaction costs, daily rebalancing could rapidly send the portfolio's wealth to zero. The investor attempting to approximate the Universal portfolio may wish to rebalance less often. This is a significant deviation from the analysis that led us to asymptotic optimality and thus we expect a large drop in performance. Furthermore, the clairvoyant investor could perform even better by tactically choosing when to rebalance based on foreknowledge of the market's behaviour. In §5.1, the effect of varying the frequency of rebalancing is investigated.

5.1 Case Study

We test the Monte Carlo Universal portfolio with $N = 10^5$ portfolio managers and $m = 2$ stocks. Ford Motor Company (F) and Bank of America Corporation (BAC) were chosen for their longevity, volatility and lack of correlation. Moreover, these stocks have similar magnitudes of return so the optimal portfolios won't be drawn to just one. In the hopes of attaining the asymptotic promises of the Universal portfolio, we first consider the 40-year period between 1985 and 2025 with no transaction costs and daily rebalancing.

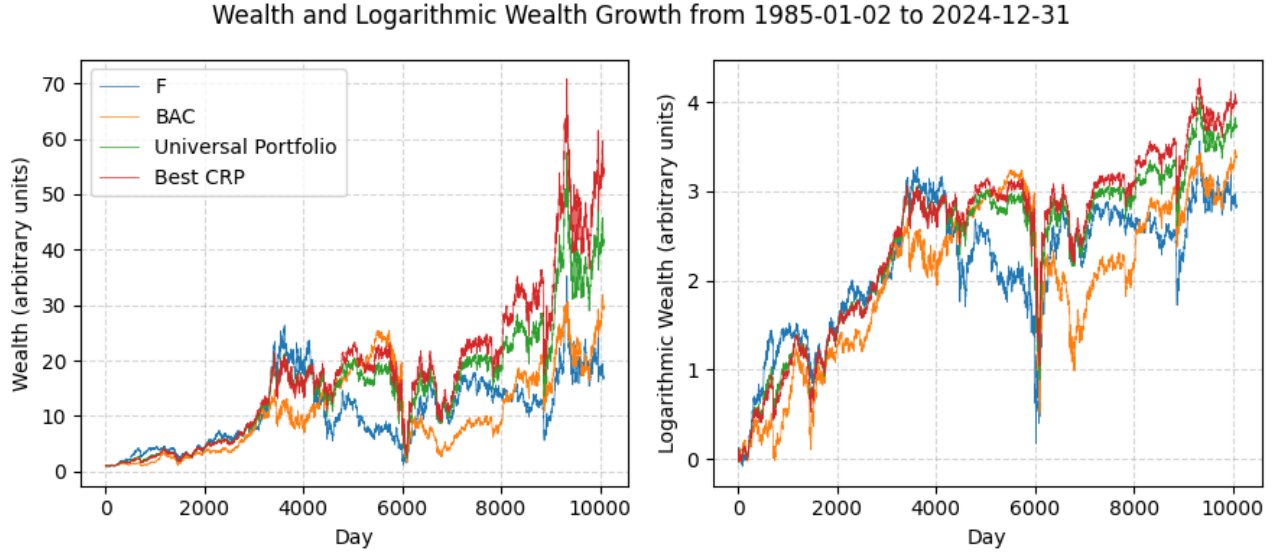


Figure 1: The variation of wealth and logarithmic wealth with time for the Monte Carlo Universal portfolio, the best constant rebalanced portfolio (CRP) chosen with hindsight and the buy and hold strategies for Ford and Bank of America. All strategies begin with unit wealth.

After this period of investment, the Monte Carlo Universal portfolio increases its wealth by a factor of 41.3 and the best possible constant rebalanced portfolio by 53.9⁵. In comparison, the best of the two underlying stocks increases in value by a factor of 29.5. Figure 1 clearly displays the approach of the Monte Carlo Universal portfolio to the benchmark. The adapting of the Monte Carlo Universal portfolio to profit off the success of each stock is evident in Figure 2. For instance, the switch from the Bank of America to Ford during the 2008 Financial Crisis allows the portfolio to recover from the crash faster than both stocks.

Consider the effect of increasing transaction costs, via c , while maintaining the daily rebalancing of the Universal portfolio. Naturally, the wealth achieved decreases with increasing c . Figure 3 shows that a commission of 0.1%, or $c = 0.001$, has minimal effect on performance even when rebalancing this often and over such a long period of time. A 10% commission, however, tends to outweigh the daily growth of the portfolio and wealth deteriorates rapidly.

⁵These values are to 3 significant figures and will differ slightly from run to run. This analysis and these plots were produced in Python using code that can be found on my [GitHub](#). Stock market data is accessed via Yahoo Finance.

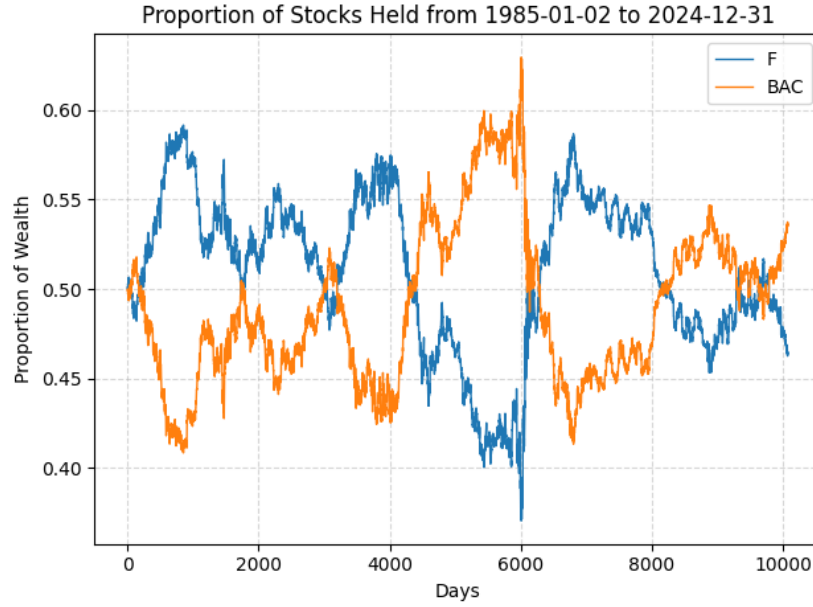


Figure 2: The proportion of the Monte Carlo Universal portfolio's wealth invested in each stock over time.

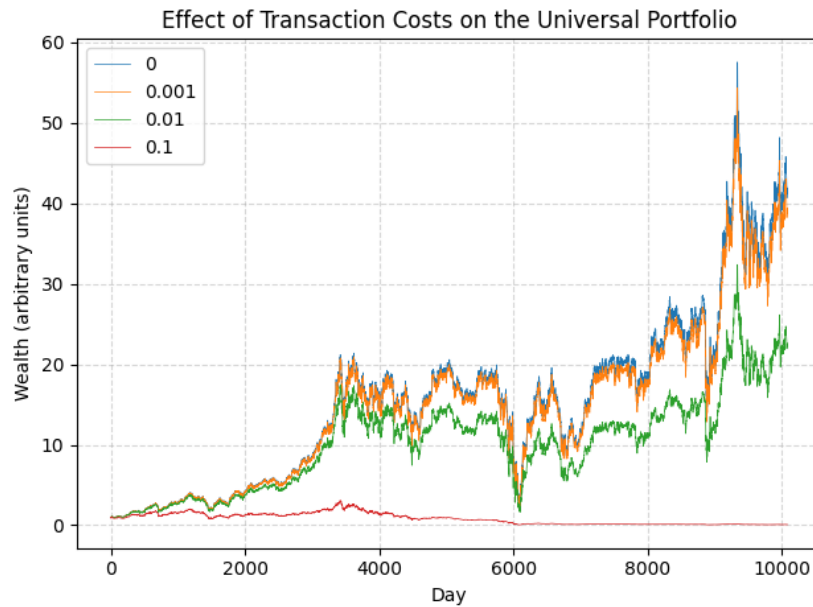


Figure 3: The variation of the Monte Carlo Universal portfolio's wealth with time for different transaction cost proportionality constants, c . All portfolios are rebalanced daily.

The effects of varying rebalancing frequencies with transaction costs present, $c \neq 0$, are less trivial. Initially, rebalancing less often has limited impact on the optimality of the Universal portfolio but significantly lowers transaction costs. However, frequencies too large stray too far from the Universal portfolio and the wealth suffers, despite the wealth saved on transaction costs. This is evident in Figure 4.

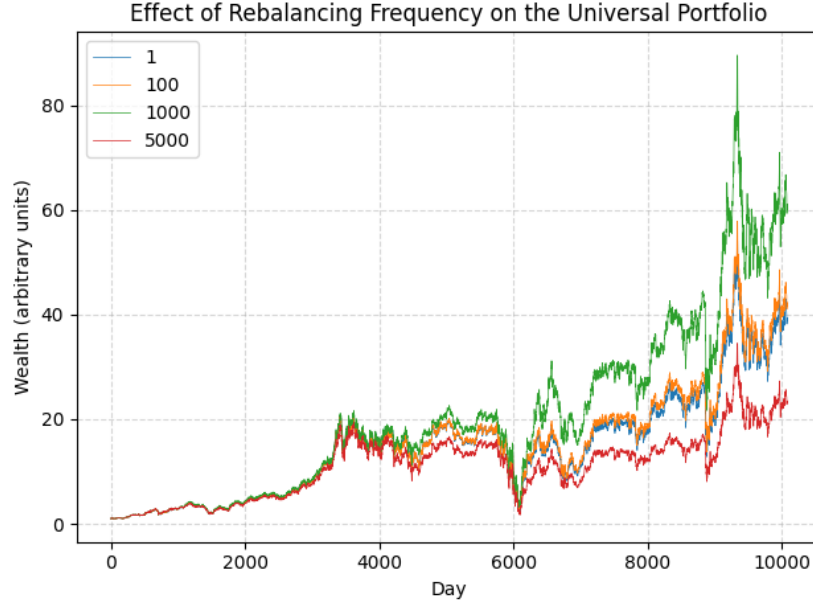


Figure 4: The variation of the Monte Carlo Universal portfolio's wealth with time for different portfolio rebalancing frequencies (in units of days). All portfolios subject to a 0.1% commission corresponding to $c = 0.001$.

An investor has no control over the commission the transactions are subject to but can select the rebalancing frequency based on the context of the problem. The decision is a trade-off between better approximating the Universal portfolio and spending less on transaction costs. Retail investors typically face larger transaction costs than institutional investors, suggesting they should rebalance their portfolios less often.

Instead of a fixed rebalancing frequency, Cover [15] put forward a potential rebalancing condition dependent on the increase in growth rate from the rebalancing outweighing the logarithm of the transaction cost to do so.

6 Duality between Investment and Data Compression

The deep connection between the fields of portfolio selection and information theory was first uncovered by Kelly [7] and has since been explored further by Cover [8, 20, 21, 25] in particular.

Say we have a finite alphabet of z possible symbols: $\mathcal{A} := \{s_1, \dots, s_z\}$. A data compression scheme aims to describe this alphabet in as few bits as possible by encoding each symbol s with $l(s)$ bits. Let X be a random symbol in $\mathcal{A}$ with probabilities $p(s) := \mathbb{P}(X = s)$. Shannon code assigns more bits to less frequent symbols to minimise the expected code length⁶.

$$l(s) = -\log(p(s)) \tag{6.1}$$

⁶Equation (6.1) is the ideal Shannon code length in the sense that it need not output an integer number of bits. These properties may also be shown for the realistic Shannon code which is the ceiling function of this length. A dyadic distribution, in which $-\log(p(s))$ is an integer, may use this ideal Shannon code.

Under Shannon code, the expected code length of this random symbol, X , in bits is exactly the Shannon entropy, $H(X)$ [22].

$$\mathbb{E}[l(X)] = - \sum_{i=1}^z p(x_i) \log(p(x_i)) \quad (6.2)$$

$$-\mathbb{E}[\log(p(X))] = H(X) \quad (6.3)$$

The data compressor aims to minimise the expected code length without losing any information. Just as the log-optimal investor maximises the expected log return, the Shannon data compressor maximises the expected log probability by using the true distribution, $p(s)$. Comparing (6.3) with (3.4), the doubling rate, W , seems to hold the same role as the negative entropy, $-H$. It turns out that the attractive properties of the log-optimal portfolio are dual to existing properties of Shannon code.

Let $X_1, \dots, X_n$ be a sequence of i.i.d. random symbols drawn from $\mathcal{A}$. The strong law of large numbers provides an equivalent form of (3.3) here.

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} -\frac{1}{n} \log(p(X_1, \dots, X_n)) = H(X) \right) = 1 \quad (6.4)$$

In other words, Shannon code is asymptotically optimal in the sense that the realised average code length approaches the minimum expected code length, the entropy, as $n \rightarrow \infty$. This is the asymptotic equipartition property for an i.i.d. source of symbols. As in the logarithmic investment counterpart, one can show a similar result for stationary, ergodic sources which is known as the Shannon-McMillan-Breiman theorem [8].

Cover [23] proved that Shannon code is also competitively optimal against other compression schemes. Let $l'(s)$ be the code length of each symbol s using any other coding strategy. Assuming the true distribution is dyadic, then this property is expressed as:

$$\mathbb{P}(l(X) < l'(X)) > \mathbb{P}(l(X) > l'(X)). \quad (6.5)$$

Therefore, Shannon code is more likely to have a shorter length, for a given symbol, than any other scheme than have a longer length.

Finally, in an exact parallel with logarithmic investment, universal data compression schemes exist which perform asymptotically as well as if the distribution of symbols was known. One such example is provided by Lempel and Ziv [24] and is the cornerstone of the algorithms behind many file formats today.

7 Conclusion

The theoretical advantages of the log-optimal portfolio should be very evident by now. We have shown investing both in the long term and short term is optimal under this choice of portfolio. Uncertainties in the benefits of this approach to portfolio selection only arise when the underlying assumptions are weakened and we try to adapt the method for real-world usage. Despite this, in the presence of realistic transaction costs and real-world data, the Universal portfolio was shown to perform significantly better than any of the underlying stocks in our case study, §5.1. Further work could focus on the implementation of a rebalancing decision condition. This would allow the investor to save on transaction costs by only rebalancing to the correct universal portfolio when the decision to do so is accompanied by a sufficiently large increase in wealth.

This analysis has been put forward in such a way as to highlight the parallels between investment in the

stock market and data compression. The optimal strategy for data compression, Shannon code, has a fundamental lower bound of H on the realised code length while the log-optimal portfolio has a fundamental upper bound of W^* on the realised growth rate. Just as with the problem of data compression, we find an asymptotic equipartition property, a competitive optimality property and a universal algorithm for the problem of investment. Real-world applications of universal data compression algorithms today provide hope that this equivalency may imply universal portfolios can be viable strategies.

References

- [1] Markowitz, H. (1952) *Portfolio Selection*, The Journal of Finance, Vol. 7 No. 1 p77-91.
- [2] Boyd, S. and Vandenberghe, L. (2004) *Convex Optimisation*, Cambridge University Press, 4.7.6 p185-187.
- [3] Mandelbrot, B. (1963) *The Variation of Certain Speculative Prices*, The Journal of Business, Vol. 36 No. 4 p394-419.
- [4] Fama, E.F. (1965) *The Behaviour of Stock-Market Prices*, Journal of Business, Vol. 38 Issue 1 p34-105.
- [5] Bell, R.M. and Cover, T.M. (1980) *Competitive Optimality of Logarithmic Investment*, Mathematics of Operations Research, Vol. 5 No. 2 p161-166.
- [6] Hanoch, G. and Levy, H. (1969) *The Efficiency Analysis of Choices Involving Risk*, The Review of Economic Studies, Vol. 36 No. 3 p335-346.
- [7] Kelly, J.L. (1956) *A New Interpretation of Information Rate*, Bell System Technical Journal, p917-926.
- [8] Cover, T.M. and Thomas, J.A. (1991) *Elements of Information Theory*, Chapter 15: Information Theory and the Stock Market, p459-481.
- [9] Samuelson, P.A. (1971) *The "Fallacy" of Maximizing the Geometric Mean in Long Sequences of Investing or Gambling*, Proceedings of the National Academy of Sciences, Vol. 68 No. 10 p2493-2496.
- [10] Thorp, E.O. (1971) *Portfolio Choice and the Kelly Criterion*, Proceedings of the Business and Economics Section of the American Statistical Association, p215-224.
- [11] Breiman, L. (1961) *Optimal Gambling Systems for Favorable Games*, Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Vol. 4 p65-78.
- [12] Bell, R.M. and Cover, T.M. (1988) *Game-Theoretic Optimal Portfolios*, Management Science, Vol. 34 No. 6 p724-733.
- [13] Algoet, P.H. and Cover, T.M. (1988) *Asymptotic Optimality and Asymptotic Equipartition Properties of Log-Optimum Investment*, The Annals of Probability, Vol. 16 No. 2 p876-898.
- [14] Cover, T.M. (1984) *An Algorithm for Maximizing Expected Logarithm Investment*, IEEE Transactions on Information Theory, Vol. 30 No. 2 p369-373.
- [15] Cover, T.M. (1991) *Universal Portfolios*, Mathematical Finance, Vol. 1 No. 1 p1-29.
- [16] Cover, T.M. and Gluss, D.H. (1986) *Empirical Bayes Stock Market Portfolios*, Advances in Applied Mathematics, Vol. 7 Issue 2 p170-181.
- [17] Robbins, H. (1956) *An Empirical Bayes Approach to Statistics*, Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Vol. 1 p157-163.
- [18] Blackwell, D. (1956) *An Analog of the Minimax Theorem for Vector Payoffs*, Pacific Journal of Mathematics, Vol. 6 No. 1 p1-8.
- [19] Blum, A. and Kalai, A. (1999) *Universal Portfolios With and Without Transaction Costs*, Machine Learning, Vol. 35 p193-205.

- [20] Cover, T.M. (1996) *Universal Data Compression and Portfolio Selection*, Proceedings of 37th Conference on Foundations of Computer Science, p534-538.
- [21] Cover, T.M. and Ordentlich, E. (1996) *Universal Portfolios with Side Information*, IEEE Transactions on Information Theory, Vol. 42 No. 2 p348-363.
- [22] Shannon, C.E. (1948) *A Mathematical Theory of Communication*, The Bell System Technical Journal, Vol. 27 Issue 3 p379-423.
- [23] Cover, T.M (1999) *On the Competitive Optimality of Huffman Codes*, IEEE Transactions on Information Theory, Vol. 31 No. 1 p172-174.
- [24] Lempel, A. and Ziv, J. (1977) *A Universal Algorithm for Sequential Data Compression*, IEEE Transactions on Information Theory, Vol. 23 No. 3 p337-343.
- [25] Cover, T.M. (1998) *Shannon and Investment*, IEEE Information Theory Society Newsletter, Special Golden Jubilee Issue, p10-11.
- [26] Williams, J.B. (1938) *Speculation and the Carryover*, The Quarterly Journal of Economics, Vol. 50 No. 3 p436-455.
- [27] Latané, H.A. (1959) *Criteria for Choice Among Risky Ventures*, Journal of Political Economy, Vol. 67 No. 2 p144-155.
- [28] Cover, T.M. and Iyengar, G.N. (2000) *Growth Optimal Investment in Horse Race Markets with Costs*, IEEE Transactions on Information Theory, Vol. 46 Issue 7 p2675-2683.
- [29] Iyengar, G.N. (2005) *Universal Investment in Markets with Transaction Costs*, Mathematical Finance, Vol. 15 Issue 2 p359-371.